

# Supplementary Information

Uncertainty-aware haplotype phasing of genetic and epigenetic variation from long reads

Jyun-Hong Lin, Yang-Ming Yeh, Yu-Chun Chen, Yao-Ting Huang

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## Supplementary Methods

### Supplementary Method 1. Allele-specific methylation calling (`modcall`)

**Modification interpretation.** Only the 5mC probability (modification code `m`) was used. 5hmC calls (code `h`) were ignored; under a 5mCG\_5hmCG model, a cytosine called as 5hmC was counted as unmethylated whenever its 5mC probability was  $\leq 0.2$ . Reverse-strand calls were mapped to the forward-strand CpG coordinate, and the two strands were genotyped jointly. Non-CpG modification calls were discarded.

**Linking modes and the depth bound on the neighbour scan.** In *SNV-anchored* mode, used whenever an SNV VCF was supplied, a candidate CpG was accepted as a strong anchor if it satisfied the co-segregation criterion of Methods against an SNV among the next 20 variants (`-connectAdjacent`). In *methylation-only* mode, used where no informative SNV was in range, the same criterion was applied between two candidate CpGs. In both modes a pair was considered only while the number of reads spanning it exceeded  $\max(6, (n_1 + n_2)/4)$ , where  $n_1$  and  $n_2$  are the numbers of reads recorded at the two sites. The neighbour scan for a site stopped at the first pair at or below this bound. Candidates that never satisfied the co-segregation criterion were retained as weak candidates. They were admitted if they satisfied the criterion on at least six reads against an already accepted CpG, in two expansion rounds (`-iterCount`, default 2).

**Merging runs of CpGs.** Runs of consecutive CpG coordinates were merged into a single locus represented by the first position, and the strand-specific observations of the run were consolidated into one read list, so that methylation loci were phased jointly with sequence variants. The resulting VCF records the methylated and unmethylated read names (`MR`, `NR`), the read strand and the per-site depths, and is passed to phase through `-mod-file`.

### Supplementary Method 2. Read-level allele observations and copy-number interval detection

**Variant eligibility and repeat annotation.** Homozygous and filtered small-variant records were carried through to the output but did not enter the phasing graph. An optional indel-quality filter excluded indels below a user-specified `QUAL` threshold and marked their `FILTER` field; the default threshold of 0 disabled this filter. An indel was flagged as a tandem-repeat indel when the dinucleotide immediately downstream of its position occurred in five consecutive copies. Flagged indels remained in the graph and cast votes at one-tenth weight. Homozygous SVs, SVs whose breakpoint coincided with an SNV and SVs with duplicated positions were excluded. The supporting read names and length of each retained SV were recorded.

**Per-class allele extraction.** SNV alleles were read from the aligned base and inherited its Phred base quality. Insertion and deletion alleles were called from the CIGAR string: a read supports the ALT allele if a CIGAR insertion or deletion begins immediately after the variant position, and the REF allele otherwise; when a read deletion spans an SNV in a homopolymer of length  $\geq 3$ , the SNV allele was recovered from the base following the deletion. SV alleles were assigned by scanning a window of 20 CIGAR operations (`-svWindow`) around the SV start for an insertion or deletion whose length is within 10% of the SV length (`-svThreshold`); reads listed by the SV caller as supporting the SV were also assigned the ALT allele. 5mC alleles were taken from the `modcall` read lists (methylated = allele 0, unmethylated = allele 1) restricted to reads whose strand matched the recorded strand. Soft- and hard-clipping operations longer than 5 bp were counted per reference position, separately for clips at the front (left) and back (right) end of an alignment, as evidence of alignment breakpoints for copy-number detection (below).

The copy-number-aware filter (Methods; Supplementary Fig. 1e,f) requires the boundaries of segments that recruit reads from more than one locus. These boundaries were detected from clusters

of soft- and hard-clips by a scan of the per-position clipping counts of each chromosome in coordinate order.

**Input.** For every reference position at which at least one alignment carries a clipping operation longer than 5 bp, two counts are recorded:  $u$ , the number of alignments clipped at their front (left) end, and  $d$ , the number clipped at their back (right) end. An interval still open at the end of the chromosome was evaluated at a position 30 kb beyond the last observed clip.

**State.** The scanner is in one of three states: *idle*, *step* (a copy-number transition, entered on a strong front-clip cluster) and *ramp* (the progressively nested breakpoints left by breakage–fusion–bridge cycles, entered on a weak front-clip excess). In the two active states it carries a running excess  $c$  of front- over back-clips, a candidate start, a candidate end and three adaptive thresholds derived from the front-clip count  $n$  that opened or re-anchored the interval:

$$r = n, \quad p = \begin{cases} \lfloor n/2 \rfloor, & n \geq 10 \\ 5, & n < 10 \end{cases},$$

$$f = \begin{cases} 5, & n \geq 20 \\ \lfloor n/4 \rfloor, & 10 \leq n < 20 \\ 2, & n < 10, \end{cases}$$

where  $r$  is the re-anchor threshold,  $p$  the pull-down threshold and  $f$  the decay floor.

**Transitions.** With constants  $\Pi = 5$  (push threshold),  $A = 30,000$  bp (look-ahead),  $E = 30$  (extend threshold),  $\Lambda = 200,000$  bp (maximum interval length) and  $\Sigma = 20$  (ramp commit threshold), the scanner at position  $x$  with counts  $(u, d)$  behaves as follows:

```

idle:
  if u >= PI and c == 0:
    enter STEP; c = u - d; start = x; end = x + A
    set r, p, f from u
  elif u > d and c == 0:
    enter RAMP; c = u - d; start = x; end = x + A

step:
  if u > r:                                # stronger breakpoint: re-anchor
    start = x; end = x + A; set r, p, f from u
  c += u - d
  if c > E: end = x + A
  if d >= p:                                # pull-down reached
    emit [start, x]; reset to idle
  elif c <= f and x <= end:                 # decayed to the floor in range
    emit [start, x]; reset to idle
  if x > end or c <= 0 or x - start >= LAMBDA:
    reset to idle                          # candidate discarded

ramp:
  if (c > SIGMA and d >= c/4) or (c <= SIGMA and d >= PI):
    emit [start, x]; reset to idle
  elif u >= PI:                             # promote to STEP
    c = u - d; start = x; end = x + A; set r, p, f from u
  else:
    c += u - d
    if c > E: end = x + A
    if x > end or c <= 0 or x - start >= LAMBDA:
      reset to idle                        # candidate discarded

```

A stronger front-clip cluster re-anchored a *step* interval: the candidate start moved to the new position and the three thresholds were recomputed, so that the reported interval began at the strongest breakpoint. The decay-floor exit applied only within the look-ahead window ( $x \leq \text{end}$ ); candidates outside this window were discarded.

**Site removal.** Within each emitted interval the mismatch-rate statistic  $\text{MR}(v)$  defined in Methods is computed and heterozygous variants with  $\text{MR}(v) \geq 0.7$  are removed from all reads. A variant is scored only when both allele populations have a non-zero mean load; a site at which one population carries no other alternate allele is not removed on this basis.

**Rationale for the mismatch-load statistic.** Comparable alternate-allele loads in the two read groups give MR near 0.5. Larger loads in the reads that carry the alternate allele increase MR and may indicate reads misaligned from paralogous sequence.

[State how the five scanner constants were chosen (a sweep, on which data, or by inspection); none is currently justified.]

### Supplementary Method 3. Tuning of the voting and read-filtering parameters

The constants in this section were set empirically. Writing the ambiguity of a pair as  $s = \min(P, Q) / \max(P, Q)$ , which ranges from 0 for unanimous support to 1 for an even split, the voting rule has three levels: a pair votes at weight  $w = 20$  when it is near-certain, that is, when  $s \leq \alpha = 0.1$  with total support  $\geq 1$  or, irrespective of  $s$ , when one pairing has accumulated less than 1 unit of support while the other has at least 1. The pair votes at unit weight otherwise, and does not vote at all when  $s > \beta = 0.7$ . Each of  $\alpha$ ,  $\beta$ ,  $w$ , the base-quality threshold  $\theta = 12$  and the single-read guard parameters ( $N = 3$  votes, ambiguity cut  $\mu = 0.2$ ) was fixed by a sweep on HG002 at 10–60 $\times$  before the experiments reported here, and none was re-tuned afterwards. The sweeps were run during the development of LongPhase in 2023–2024 on whole-genome phasing of earlier HG002 nanopore alignments, before the alignments evaluated here were generated and before the chromosome split used for GNN training was defined. The parameters were therefore tuned on data from the same individual as the evaluation, and chr17, chr21 and chr22 are held out from GNN training but not from this tuning. [Supplementary Fig. 9 must either be produced or removed before submission.]

### Supplementary Method 4. Graph neural network: features, architecture, labels, training and export

**Node features.** Each allele node carried 31 features in four groups, phasing evidence, genomic context, graph structure and variant class, defined in Supplementary Table 1 (window construction in Supplementary Figs. 4 and 5). SNVs and indels were distinguished by  $||\text{ALT}| - |\text{REF}|| > 0$ .

**Edge features.** Edges were taken from the DOT graph, restricted to pairs both of whose end-points lie inside the window, symmetrized by copying the features of whichever direction was present, and augmented with a self-loop per node whose features are the mean of its non-self incident edges. The vote weight  $w$  of an edge is the weighted haplotype vote recorded for the source allele when the vote was cast. Each edge carried six features (Supplementary Table 2).

**Architecture.** The network used the GPS graph-transformer architecture<sup>1</sup> (Supplementary Table 3 and Supplementary Fig. 6). Node features were standardized with a batch-normalization layer whose statistics are frozen at inference, linearly projected to a 128-dimensional embedding and passed through a GELU non-linearity. Four identical layers then each combine a local and a global branch. The local branch is a four-head GATv2 message-passing layer<sup>2,3</sup> in which the attention

logit of head  $t$  for edge  $j \rightarrow i$  is

$$\alpha_{ij}^{(t)} \propto \exp\left(\mathbf{a}_t^\top \text{LeakyReLU}_{0.2}([\mathbf{W}_r \mathbf{x}_i + \mathbf{W}_l \mathbf{x}_j + \mathbf{W}_e \mathbf{e}_{ij}]_t)\right), \quad (1)$$

where  $[\cdot]_t$  selects the 32 coordinates of head  $t$ ,  $\mathbf{W}_e$  is a per-layer edge encoder without bias and the normalization is a softmax over the incoming edges of  $i$  (a node with no incoming edge instead receives the uniform average of  $\mathbf{W}_l \mathbf{x}_j$  over all window nodes); the head outputs are concatenated and a bias is added. The global branch is four-head scaled dot-product self-attention over all nodes of the window<sup>4</sup>, with no masking. The two branch outputs are added to the residual stream, followed by layer normalization, a two-layer feed-forward block (128→256→128, GELU) with a second residual connection and layer normalization. The final node embedding is concatenated with two edge-context scalars, the sum and the non-negative maximum, over the node’s incoming edges excluding its self-loop, of the first edge feature, i.e. of the logistic-transformed vote weight  $1/(1 + e^{-w})$  rather than of  $w$  itself. The pooled vector is then passed through a two-layer classifier (130→128→2) with softmax output.

The model comprises 687,358 stored values, including the frozen batch-normalization statistics. Inference was implemented in C++ without external runtime dependencies and used the erf-based GELU. Predictions for the two allele nodes of a variant were averaged. Only variants in the central zone of each window were scored: those whose genomic distance from the trigger is at most half the largest such distance in the window,  $|p - p_c| \leq \frac{1}{2} \max_u |p_u - p_c|$ . Only labelled variants in this zone contributed to the training loss, and the same zone was scored at inference; the other variants supplied graph context. A variant covered by several windows received the running mean of its predictions.

**Phase-set update.** A variant with a mean phase-error score  $\geq 0.30$  was unphased while preserving its alleles: both 0|1 and 1|0 became 0/1, and the PS tag was removed. Bridge protection was disabled in the reported runs. Connectivity was then evaluated over the surviving graph edges within each original phase set. The largest connected component retained the original PS identifier; each other component received the 1-based position of its first variant as its identifier. Corrected records were written separately for small variants, SVs and methylation sites.

**Training augmentations.** Two label-preserving augmentations were applied to training windows only, each independently with probability 0.5: a haplotype swap, which exchanges the two allele nodes of every variant and complements the HP feature, an operation the phasing problem is invariant under; and a coverage down-scaling, which multiplies  $h_1$ ,  $h_2$ , their sum, the outgoing-weight standard deviation and the raw edge weights by a factor drawn uniformly from  $[0.4, 0.7]$  and recomputes the derived logarithmic features, applied only to windows containing at least one error node.

**Labels.** Labels were derived from the v5.0q benchmark restricted to chr1–chr22 small variants, the same truth set used for evaluation, and describe the phase of a variant relative to its neighbours rather than in absolute terms. Within each window the variants were grouped by phase set; in each group the first variant present in the benchmark fixed the orientation of that group and was labelled *correct*, and every later variant was labelled *correct* if its allele-to-haplotype orientation agreed with that anchor and *disagreeing* otherwise. Maximal runs of consecutive disagreeing variants were then resolved so that a run is charged to the event that created the run, not to each member. A run of length one became *unphase* if that variant’s PE was at least 0.9 and *flip* otherwise, whereas a run of length two or more had its first variant, the switch point, relabelled *break*, and all remaining members reset to *correct*. Variants absent from the benchmark, and all SV and 5mC records, for which the benchmark carries no phase truth, were left unlabelled: they remained in the graph as context but did not contribute to the loss. The released model was trained in binary mode, in which *flip*, *unphase* and *break* are merged into a single *error* class.

**Class balance and loss.** Among the labelled central nodes of the training split, 17,812,710 (97.04%) were *correct* and 544,170 (2.96%) *error*, and 21.5% of windows contained at least one error node; the prevalence is far above the genome-wide switch-error rate because windows are opened only at high-entropy triggers and because the label is assigned relative to a window-local anchor rather than to a whole phase block. Each minibatch kept all of its error nodes and a uniform random sample of correct nodes of at most four times that number (`-ratio_corr 4`), drawn again for every minibatch. The cross-entropy on that subset was focal-weighted<sup>5</sup> by  $(1 - p_t)^\gamma$  with  $\gamma = 1.5$  and used label smoothing of 0.05.

**Model selection.** Checkpoints were selected on the validation chromosomes by a net-gain criterion. At each epoch the predicted phase-error score was thresholded at each of 81 values spanning  $[0.01, 0.99]$  and the score

$$S = R \left( 2 - \frac{1}{P} \right) \quad (2)$$

was maximized over those thresholds subject to a recall of at least 0.005, where  $P$  and  $R$  are the precision and recall of the error class;  $S$  equals the number of correctly detected errors minus the number of false detections, as a fraction of all errors, and is positive only for  $P > 0.5$ . Epochs with  $P \leq 0.5$  were instead ordered by  $P - 1$ . The selected checkpoint was epoch 44 ( $S = 0.0235$  at  $P = 0.626$ ,  $R = 0.058$  and a threshold of 0.868); its classification metrics on the validation and held-out chromosomes are given in Supplementary Table 4. The score  $S$  governed checkpoint selection; the deployment threshold was chosen separately using the end-to-end phasing trade-off described below.

**Deployment-threshold selection.** Phase-error score thresholds  $B$  from 0.05 to 0.95 in steps of 0.05 were evaluated on 60 HG002 nanopore runs, comprising ten down-sampling replicates at each of 10, 12, 14, 16, 18 and 20 $\times$ . Each threshold was scored, relative to phasing without GNN correction, by the fraction of switch errors removed and by the cost of correction, the number of phased SNVs withheld divided by the number of switch errors removed. Because the two trade off against each other, the criterion was the largest fraction of switch errors removed at a cost of at most ten withheld SNVs per removed error. For the released model,  $B = 0.30$  removed 33.1% of switch errors at a cost of 8.3 withheld SNVs per removed error (60 runs pooled, whole genome). On the held-out chromosomes chr17, chr21 and chr22, the same threshold removed 21.2% of switch errors at a cost of 10.5, slightly above the limit; the threshold was not re-selected on these chromosomes (Supplementary Table 15). This criterion concerns the final phasing output and is distinct from the classifier-based criterion used to rank training checkpoints.

**Export and numerical agreement.** The trained model was rewritten with dense  $N \times N$  adjacency and edge tensors, exported to ONNX (opset 17, dynamic  $N$ ) and compiled into the C++ module; the logistic transform of the vote weight and the self-loop features are applied to the inputs before inference. Agreement was checked at each step: the dense rewrite against the PyTorch Geometric model and the ONNX graph against the dense rewrite, on a real training window, with an acceptance bound of  $10^{-3}$  on the absolute difference in phase-error score; and the compiled C++ module against ONNX Runtime on six synthetic graphs of 3–120 nodes at edge densities 0.1–0.4, where the largest absolute difference was  $3 \times 10^{-7}$  (acceptance bound  $10^{-5}$ ).

## Supplementary Method 5. Read haplotagging

`longphase haplotag` assigns each alignment to a haplotype using the phased VCFs produced by phasing with or without GNN correction.

**Alignment selection.** Primary alignments with mapping quality  $\geq 1$  (`-q`, `-qualityThreshold`) are considered. Supplementary alignments are tagged only under `-tagSupplementary`; secondary

alignments never are. Any pre-existing HP, PS and PQ tags are removed from every record before tagging. Unmapped records are passed through unchanged.

**Evidence collection.** Each alignment is traversed once along its CIGAR, and four classes of evidence are collected into two per-haplotype tallies ( $n_1, n_2$ ):

- *SNVs* are read from M, = and X operations; a read carrying the allele phased to haplotype  $k$  adds 1 to  $n_k$ .
- *Small indels* are scored by allele length rather than by sequence comparison. Writing  $\ell_1$  and  $\ell_2$  for the lengths of the two phased alleles at the site, a read that carries an indel there supports the haplotype whose allele has  $\ell \neq 1$ , and a read that does not supports the haplotype whose allele has  $\ell = 1$ . When both or neither allele has length 1 the site casts no vote.
- *SVs* are re-derived from the alignment, not taken from the caller’s read list: a window of  $\pm$ -svWindow CIGAR operations (default 20) around the current operation is scanned for an insertion or deletion whose length is within -svThreshold (default 0.10) of the SV length, and the first match adds 1 to the haplotype the SV is phased to.
- *Allele-specific 5mC* sites are matched by read name, not by position: the methylated-read list INFO/MR of the phased 5mC VCF adds 1 to the haplotype that record is phased to, for each read it names. The unmethylated list NR is not consulted; a 5mC site contributes evidence only to the reads that are methylated at it.

Each informative observation contributed unit weight during haplotagging, whatever its class. During read-based correction inside **phase**, by contrast, SNV and SV alleles contribute 1, indel and tandem-repeat-indel alleles 0.1, and 5mC alleles are excluded (Methods).

**Decision and phasing quality.** Let  $m = \max(n_1, n_2)$  and  $\mu = \min(n_1, n_2)$ . The read is assigned to the majority haplotype when  $m/(m + \mu) \geq -p$ , -percentageThreshold (default 0.6) and is otherwise left untagged. A Phred-scaled phasing quality is computed for every alignment as

$$PQ = \begin{cases} 0 & m = 0, \\ 40 & \mu = 0, \\ \left\lfloor -10 \log_{10} \frac{\mu}{m + \mu} \right\rfloor & \text{otherwise,} \end{cases} \quad (3)$$

PQ is written as an integer auxiliary tag PQ:i alongside HP:i and PS:i, and only on tagged reads.

**Phase-set veto.** The phase sets of all informative variants on the read are counted. Reads with informative variants in more than one phase set were left untagged, regardless of their haplotype tally. The PS value written on a tagged read is that of the first phased variant at or after the alignment start, whether or not that variant is informative on the read; because the veto has already required every informative variant to share one phase set, the two coincide except when that first variant is uninformative on the read and belongs to a different block, a boundary case.

## Supplementary Method 6. Evaluation metrics

Metrics were computed with `longphase compare` (Supplementary Fig. 8), a C++ re-implementation of the `whatshap compare` algorithm<sup>6</sup>.

**Block intersection.** For each chromosome, heterozygous records common to the truth and a query are intersected by position and by REF/ALT alleles. The intersection is partitioned into maximal runs that lie in one truth phase set and one query phase set simultaneously; runs of length one carry no phase information and are discarded. The switch errors and the Hamming distance

of a query are computed inside its surviving blocks but with different denominators: the switch error rate divides SNV switch errors by the number of consecutive SNV pairs assessed, whereas the Hamming distance divides misassigned variants, of every class present in both files, by the number of compared variants; the phased fraction is counted before this partition, as the number of truth heterozygous variants whose position, REF and first ALT match a query record that is heterozygous and phased, over all truth heterozygous variants of the class.

**Switch errors and flips.** Each block yields two binary strings, the truth haplotype assignment  $p_0$  and the query assignment  $q_0$  of its variants. Both are converted to *switch encoding*, the string of relative changes between consecutive positions, which removes the arbitrary orientation of the block. Switch errors are the Hamming distance between the two switch-encoded strings. The decomposition into isolated switches and flips follows WhatsHap’s `compute_switch_flips`: a maximal run of  $k$  consecutive disagreements in switch space is charged as  $\lfloor k/2 \rfloor$  flips and  $k \bmod 2$  switches. A single variant assigned to the wrong haplotype thus counts as one flip, not two switches. The *switch error rate* is the total number of switch errors divided by the number of assessed consecutive pairs.

**Hamming distance.** For each block the query strings  $(q_0, q_1)$  are compared with the truth strings  $(t_0, t_1)$  in both orientations, and the smaller of  $H(t_0, q_0) + H(t_1, q_1)$  and  $H(t_0, q_1) + H(t_1, q_0)$  is halved to give the number of misassigned variants in that block. Block values are summed and expressed as a percentage of compared variants.

**Block N50.** Block spans in bp are sorted in decreasing order and the N50 is the span at which the cumulative sum first reaches half the total, which is the span-weighted median. It is computed twice: once per chromosome, and once over the block spans of all chromosomes pooled. The values plotted in Figs. 2c,g,k, 3b,c and 5c are the pooled, genome-wide N50 of the query, which does not depend on the truth set.

## Supplementary Tables

**Supplementary Table 1: Node features of the graph neural network.** Each allele node of a window carries 31 features; # is the 1-based index in the feature vector, listed here by group, not by index.  $h_1$ ,  $h_2$ : weighted haplotype votes (taken from the DOT edge labels when available, otherwise from INFO/H1, INFO/H2); PE: phasing entropy; PS: phase set; “centre” is the trigger variant of the window. Genomic-context features are computed from the reference when -r is given and are zero otherwise.

| #                                 | Feature                         | Definition                                                                              |
|-----------------------------------|---------------------------------|-----------------------------------------------------------------------------------------|
| <i>Phasing evidence (16)</i>      |                                 |                                                                                         |
| 1                                 | PE                              | normalized Shannon entropy of $(h_1, h_2)$ , 0–1 bit                                    |
| 2                                 | HP                              | haplotype of this allele node (genotype index)                                          |
| 3                                 | gt_ref                          | haplotype index of the reference allele: first digit of the phased GT (0 or 1)          |
| 4                                 | gt_alt                          | haplotype index of the alternate allele: second digit of the phased GT (0 or 1)         |
| 5                                 | is_center                       | 1 if the variant is the window trigger                                                  |
| 6                                 | is_same_ps                      | 1 if the variant shares the trigger’s PS                                                |
| 7                                 | allele                          | 0 for the reference-allele node, 1 for the alternate-allele node                        |
| 8                                 | rel_offset                      | $(\text{pos} - \text{pos}_c) / \max \text{pos} - \text{pos}_c $ over the window, signed |
| 9                                 | H1                              | $h_1$                                                                                   |
| 10                                | H2                              | $h_2$                                                                                   |
| 11                                | total_votes                     | $h_1 + h_2$                                                                             |
| 12                                | vote_ratio                      | $\max(h_1, h_2) / (h_1 + h_2)$                                                          |
| 13                                | log_total                       | $\log_2(1 + h_1 + h_2)$                                                                 |
| 14                                | log_h1                          | $\ln(1 + h_1)$                                                                          |
| 15                                | log_h2                          | $\ln(1 + h_2)$                                                                          |
| 25                                | rel_vote_depth                  | $\log_2((h_1 + h_2) / \overline{(h_1 + h_2)}_{\pm 10 \text{ variants}})$                |
| <i>Genomic context (6)</i>        |                                 |                                                                                         |
| 16                                | hp_len                          | homopolymer run length at the variant (capped at 20)                                    |
| 17                                | gc_content                      | GC fraction in $\pm 50$ bp                                                              |
| 18                                | str_context                     | longest tandem-repeat unit count (unit 2–4 bp) in $\pm 15$ bp, /10, capped at 1         |
| 19                                | seq_entropy                     | Shannon entropy of the $\pm 15$ bp sequence (bits)                                      |
| 20                                | neigh_dens                      | $(\text{number of variants within } \pm 500 \text{ bp} - 1) / 20$                       |
| 24                                | dist_near_indel                 | $\log_{10}(1 + \text{distance to nearest indel})$ , capped at $\log_{10}(10001)$        |
| <i>Graph structure (5)</i>        |                                 |                                                                                         |
| 21                                | out_degree                      | number of distinct target positions of outgoing DOT edges, /10, capped at 1             |
| 22                                | ps_block_size                   | $\ln(1 + \text{variants in the PS}) / \ln(101)$                                         |
| 23                                | log_center_dist                 | $\log_{10}(1 +  \text{pos} - \text{pos}_c )$                                            |
| 26                                | out_weight_std                  | s.d. of outgoing DOT edge weights, /100, capped at 1                                    |
| 27                                | is_bridge                       | 1 if removing the variant disconnects the window graph                                  |
| <i>Variant class (4, one-hot)</i> |                                 |                                                                                         |
| 28–31                             | is_SNV, is_indel, is_SV, is_5mC | variant class indicators                                                                |

**Supplementary Table 2: Edge features of the graph neural network.** Edges are taken from the DOT export, restricted to node pairs inside the window, symmetrized, and complemented by one self-loop per node whose features are the mean over the node’s incident edges.  $w$  is the DOT edge label (accumulated vote weight).

| # | Feature      | Definition                                                            |
|---|--------------|-----------------------------------------------------------------------|
| 1 | weight       | $1/(1 + e^{-w})$                                                      |
| 2 | log_dist     | $\log_{10}(1 +  \text{pos}_{\text{dst}} - \text{pos}_{\text{src}} )$  |
| 3 | log_weight   | $\ln(1 + w)$                                                          |
| 4 | is_same_hp   | 1 if the two allele nodes have the same allele index                  |
| 5 | is_inter_ps  | 1 if the two variants belong to different phase sets                  |
| 6 | weight_ratio | $w / (\text{sum of outgoing edge weights of the source allele node})$ |

**Supplementary Table 3: Architecture and inference hyperparameters of the graph neural network.**

| Component           | Setting                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input normalization | BatchNorm over 31 node features (frozen statistics at inference)                                                                                                    |
| Input projection    | Linear $31 \rightarrow 128$ , GELU                                                                                                                                  |
| Layers              | 4 GPS layers, hidden size 128                                                                                                                                       |
| Local branch        | GATv2, 4 heads $\times$ 32, LeakyReLU slope 0.2, per-layer edge encoder $6 \rightarrow 128$                                                                         |
| Global branch       | multi-head self-attention, 4 heads, Q/K/V/O projections $128 \rightarrow 128$                                                                                       |
| Fusion              | $x \leftarrow x + \text{local} + \text{global}$ ; LayerNorm ( $\varepsilon = 10^{-5}$ )                                                                             |
| Feed-forward        | $128 \rightarrow 256 \rightarrow 128$ , GELU, residual, LayerNorm                                                                                                   |
| Classifier          | $\text{concat}[h_i, \sum_j \sigma(w_{ij}), \max_j \sigma(w_{ij})] (130) \rightarrow 128$<br>(GELU) $\rightarrow 2$ , softmax; $\sigma$ logistic, self-loop excluded |
| Stored values       | 687,358, including frozen normalization statistics                                                                                                                  |
| Window              | trigger $\text{PE} \geq 0.8$ ; $\pm 20$ variants ( $\leq 41$ variants, $\leq 82$ allele nodes)                                                                      |
| Decision            | unphase if the mean phase-error score over the two allele nodes and all covering windows is $\geq 0.30$                                                             |
| Post-processing     | split disconnected phase sets (largest component keeps PS); bridge protection off                                                                                   |
| Training            | see Supplementary Table 4                                                                                                                                           |

**Supplementary Table 4: Training configuration and evaluation of the released model.** Values recorded in the released checkpoint and its training log. “Scored nodes” are labelled allele nodes inside the central zone of a window, the only nodes that contribute to the loss.

| Item                          | Setting                                                                                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Data</i>                   |                                                                                                                                                                                                      |
| Source runs                   | HG002 ONT R10.4.1, four-class co-phasing, 10/12/14/16/18/20 × × 10 down-sampling replicates (60 runs)                                                                                                |
| Window extraction             | -pe_threshold 0.8, -window 20, GRCh38 no-alt analysis set                                                                                                                                            |
| Truth set                     | v5.0q benchmark, chr1–chr22 small variants                                                                                                                                                           |
| Split (by chromosome)         | train chr1–chr14, chr18–chr20; validation chr15, chr16; test chr17, chr21, chr22                                                                                                                     |
| Windows / scored nodes        | train 602,788 / 18,356,880; val 67,880 / 1,509,576; test 51,448 / 1,367,580                                                                                                                          |
| Class prevalence (train)      | correct 17,812,710 (97.04%), error 544,170 (2.96%); 21.5% of windows contain an error node                                                                                                           |
| Labels                        | anchor-relative within each PS; single disagreement → <i>unphase</i> if PE ≥ 0.9 else <i>flip</i> ; run of ≥ 2 → first variant <i>break</i> , rest <i>correct</i> ; merged to one <i>error</i> class |
| Masked from loss              | variants absent from the benchmark; all SV and 5mC records (the truth set holds small variants only); nodes outside the central zone                                                                 |
| <i>Optimization</i>           |                                                                                                                                                                                                      |
| Loss                          | focal cross-entropy, $\gamma = 1.5$ , label smoothing 0.05, on a per-batch subset of all error nodes plus $\leq 4\times$ as many randomly drawn correct nodes                                        |
| Optimizer                     | AdamW, $\beta = (0.9, 0.999)$ , $\varepsilon = 10^{-8}$ , weight decay $10^{-3}$                                                                                                                     |
| Learning rate                 | $3 \times 10^{-4}$ , no warm-up, cosine annealing to $10^{-5}$ over 100 epochs                                                                                                                       |
| Batch size                    | 64 windows, batched as disjoint graphs                                                                                                                                                               |
| Regularization                | dropout 0.4 (local branch, global branch, FFN; inactive at inference); gradient-norm clipping 1.0                                                                                                    |
| Graph preprocessing           | reverse edge added per DOT edge with identical features (undirected)                                                                                                                                 |
| Augmentation (train only)     | haplotype swap, $p = 0.5$ ; coverage down-scaling $\times U[0.4, 0.7]$ , $p = 0.5$ , error-bearing windows only                                                                                      |
| Epochs                        | budget 100, early stopping after 25 epochs without improvement; stopped at 69                                                                                                                        |
| Seed                          | 42 (Python and PyTorch generators)                                                                                                                                                                   |
| Environment                   | PyTorch 2.11.0, PyTorch Geometric 2.7.0, Python 3.12, CUDA 13.0, cuDNN 9.19, one NVIDIA GeForce RTX 5090                                                                                             |
| <i>Selection and export</i>   |                                                                                                                                                                                                      |
| Selection criterion           | max over 81 thresholds in $[0.01, 0.99]$ of $R(2 - 1/P)$ subject to $R \geq 0.005$ , on chr15–chr16; $P - 1$ where $P \leq 0.5$                                                                      |
| Selected checkpoint           | epoch 44; $S = 0.0235$ at $P = 0.626$ , $R = 0.058$ , threshold 0.868                                                                                                                                |
| Held-out test (chr17, 21, 22) | argmax: accuracy 0.941, error-class $P/R/F_1 = 0.258/0.499/0.340$ ; tuned threshold: 0.946, 0.273/0.468/0.345; $n = 41,378$ error nodes                                                              |
| Deployment threshold          | 0.30; phased-SNV/switch-error trade-off (Supplementary Method 4)                                                                                                                                     |
| Export                        | ONNX opset 17, dynamic $N$ , compiled into the C++ module; C++ vs. ONNX Runtime max. absolute difference $3 \times 10^{-7}$                                                                          |

**Supplementary Table 6: Selected applications and evaluations of LongPhase.** Selected from works citing LongPhase (Europe PMC and OpenAlex, September 2026); the last column paraphrases the Methods of each work.

| Work                                                    | Journal, year                     | Relation to LongPhase                                                                                                                   |
|---------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <i>Variant callers and pipelines</i>                    |                                   |                                                                                                                                         |
| ClairS <sup>10</sup>                                    | Nat. Methods<br>2026              | default germline phasing of tumour and normal alignments ( <code>-whatshap_for_phasing</code> the alternative); haplotagging selectable |
| ClairS-TO <sup>11</sup>                                 | Nat. Commun.<br>2025              | default intermediate phasing and haplotagging of the tumour alignment (WhatsHap selectable)                                             |
| Clair3-RNA <sup>12</sup>                                | Nat. Commun.<br>2025              | phasing in the long-read RNA pipeline                                                                                                   |
| Clair3 (accelerated) <sup>13</sup>                      | Bioinformatics<br>2026            | selectable intermediate phasing and haplotagging ( <code>-longphase_for_phasing</code> ; WhatsHap the default)                          |
| Severus <sup>14</sup>                                   | Nat. Biotechnol.<br>2025          | one of the documented routes to the haplotagged BAMs it consumes (WhatsHap the other)                                                   |
| EPI2ME wf-human-variation <sup>15</sup>                 | ONT workflow                      | intermediate per-contig phasing ( <code>phase_contig</code> ); WhatsHap for final phasing and haplotagging                              |
| HiFi-somatic-WDL <sup>16</sup><br>cuteHap <sup>17</sup> | PacBio workflow<br>Adv. Sci. 2026 | optional phasing tool<br>read phasing before<br>haplotype-aware SV calling                                                              |
| SpecImmune <sup>18</sup>                                | Adv. Sci. 2026                    | joint SNV–SV phasing of immune genes                                                                                                    |
| <i>Biological and clinical studies</i>                  |                                   |                                                                                                                                         |
| Fluctuating DNA methylation in cancer <sup>19</sup>     | Nature 2025                       | read phasing (v1.7) and <code>modcall</code> for CpG methylation                                                                        |
| Advanced cancer cohort (POG) <sup>20</sup>              | Cell Genomics<br>2024             | discussed as a phasing method for tumours                                                                                               |
| Pediatric epilepsy epesignatures <sup>21</sup>          | Nat. Commun.<br>2024              | phasing of Clair3 and SV calls                                                                                                          |
| GNAS transcripts in PHP1B <sup>22</sup>                 | PNAS 2025                         | phasing of Clair3 calls                                                                                                                 |
| 3-hour genome for neonatal risk <sup>23</sup>           | Genet. Med. Open<br>2024          | real-time phasing during sequencing                                                                                                     |
| Ultrarapid diagnostics in critical care <sup>24</sup>   | Eur. J. Hum. Genet. 2025          | read phasing (v1.0.7)                                                                                                                   |
| HYDIN in primary ciliary dyskinesia <sup>25</sup>       | Eur. J. Hum. Genet. 2024          | phasing of Clair3 calls                                                                                                                 |
| Antisense oligonucleotides in APBD <sup>26</sup>        | Nucleic Acids Res.<br>2025        | phasing of Clair3 calls                                                                                                                 |
| Native CpG methylation fine-mapping <sup>27</sup>       | HGG Adv. 2026                     | phasing and haplotagging                                                                                                                |
| Canadian national platform benchmark <sup>28</sup>      | Front. Genet. 2026                | phasing inside the Clair3 workflow                                                                                                      |
| Pallas’s cat genome <sup>29</sup>                       | NAR Genom. Bioinform. 2023        | haplotagging for allele-specific methylation                                                                                            |
| <i>Methods evaluated against LongPhase</i>              |                                   |                                                                                                                                         |
| HiPhase <sup>30</sup>                                   | Bioinformatics<br>2024            | comparison method                                                                                                                       |
| Ralphi <sup>31</sup>                                    | Genome Res. 2025                  | comparison method                                                                                                                       |

| Work                  | Journal, year                 | Relation to LongPhase               |
|-----------------------|-------------------------------|-------------------------------------|
| GCphase <sup>32</sup> | BMC<br>Bioinformatics<br>2024 | comparison method                   |
| Duet <sup>33</sup>    | BMC<br>Bioinformatics<br>2022 | comparison method for SV<br>phasing |

**Supplementary Table 5: Software and data used in the benchmarks.**

| Item                                      | Version                             | Notes                                                                                                                                                                                                                                                       |
|-------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HG002 nanopore R10.4.1 reads              | ONT<br>giab_lsk114_2022.12          | FLO-PRO114M, SQK-LSK114, 4 kHz;<br>Dorado sup@v4.0.0 with 5mCG_5hmCG@v2                                                                                                                                                                                     |
| Reference                                 | GRCh38                              | GCA_000001405.15_GRCh38_no_alt_analysis_set; no-alt analysis set, no decoys                                                                                                                                                                                 |
| minimap2                                  | 2.24-r1122                          | -y -ax map-ont (alignment by Oxford Nanopore; reads with mean quality $\geq 10$ )                                                                                                                                                                           |
| PEPPER-Margin-DeepVariant <sup>7</sup>    | r0.8                                | SNV and indel calls (unphased genotypes); call_variant with -ont_r10_q20 (nanopore) or -hifi (HiFi), one call set per coverage and replicate                                                                                                                |
| Sniffles2                                 | 2.8.0                               | -output-rnames                                                                                                                                                                                                                                              |
| GIAB v4.2.1 HG002                         | v4.2.1                              | GRCh38 small-variant benchmark with phase transferred from a hifiasm assembly; HG002_GRCh38_1_22_v4.2.1_benchmark_hifiasm_v11_phasetransfer.vcf.gz, biallelic phased heterozygous records                                                                   |
| GIAB v5.0q HG002 benchmark <sup>8,9</sup> | v5.0q                               | GRCh38 small-variant benchmark derived from the T2T-HG002 v1.1 assembly; HG002_GRCh38_v5.0q_smvar.vcf.gz, chr1–chr22 records                                                                                                                                |
| WhatsHap                                  | 2.8                                 | phase -ignore-read-groups<br>-reference; -only-snvs for SNV comparisons; default indel phasing for indel comparisons                                                                                                                                        |
| HapCUT2                                   | 1.3.4                               | extractHAIRS -ont 1, then HAPCUT2 with default parameters (default pruning); input restricted to heterozygous SNVs                                                                                                                                          |
| Margin                                    | 2.3.1                               | SNV-only comparison, 10–20 $\times$ ; phasing parameters<br>allParams.phase_vcf.ont.json; input restricted to SNVs                                                                                                                                          |
| Ralphi                                    | v0.0.1 (commit d7d17b1)             | SNV-only comparison, 10–20 $\times$ ; model ralphi.v1.long.pt, ONT configuration with max_coverage 15, mapq 20, read selection and bad-read filtering                                                                                                       |
| MethPhaser                                | 0.0.4                               | run on the uncorrected SNV-only LongPhase 2 output (same SNV phasing as the joint SNV+5mC run), replicate 1<br><a href="#">[command line]</a>                                                                                                               |
| HG002 PacBio HiFi reads                   | Revio, PacBio release<br>2023-10-31 | ccs 7.0.0, lima 2.7.1, primrose 1.4.0; aligned with pbmm2 1.10.0 (48 $\times$ ); down-sampled with samtools 1.23.1 (seed 1) to 10–40 $\times$ , 48 $\times$ used as 50 $\times$ ; PEPPER-Margin-DeepVariant r0.8 -hifi, called separately at every coverage |
| LongPhase 1.0                             | v1.0                                | -ont; commit fe8c1fdff11ba2a082d363a7ffcb5c923cacc316                                                                                                                                                                                                       |
| LongPhase 2                               | v2.1, commit 46ba470                | Nanopore mode (-pb for HiFi); GNN correction with defaults                                                                                                                                                                                                  |

**Supplementary Table 7: Input call sets per coverage.** Heterozygous (het.) and homozygous (hom.) calls entering the phasing runs; nanopore values at 10–20× are means (s.d.) over ten down-sampling replicates. Small variants from PEPPER-Margin-DeepVariant, SVs from Sniffles2 2.8.0, allele-specific 5mC sites from longphase modcall. SVs and 5mC were not called on the HiFi data. Homozygous SNVs include the PEPPER candidates genotyped homozygous reference (refCall).

| Platform | Cov. | Het. SNV          | Hom. SNV          | Het. indel    | Het. SV      | Hom. SV    | Het. 5mC        |
|----------|------|-------------------|-------------------|---------------|--------------|------------|-----------------|
| Nanopore | 10×  | 2,340,598 (3,826) | 3,265,243 (8,954) | 283,612 (381) | 14,866 (130) | 7,684 (57) | 283,932 (1,026) |
| Nanopore | 12×  | 2,506,797 (2,922) | 3,227,137 (8,307) | 315,147 (511) | 16,119 (69)  | 7,874 (33) | 361,073 (2,274) |
| Nanopore | 14×  | 2,586,505 (3,184) | 3,173,111 (5,212) | 332,537 (343) | 16,707 (67)  | 8,008 (31) | 408,058 (1,847) |
| Nanopore | 16×  | 2,621,584 (2,784) | 3,115,855 (8,105) | 342,202 (505) | 17,089 (68)  | 8,121 (17) | 431,895 (976)   |
| Nanopore | 18×  | 2,636,024 (1,840) | 3,051,563 (5,197) | 347,703 (324) | 17,326 (80)  | 8,201 (34) | 444,081 (1,158) |
| Nanopore | 20×  | 2,640,440 (2,408) | 3,004,806 (5,806) | 350,682 (328) | 17,487 (43)  | 8,247 (32) | 452,032 (1,079) |
| Nanopore | 30×  | 2,636,000         | 2,902,347         | 356,240       | 17,800       | 8,473      | 474,512         |
| Nanopore | 40×  | 2,617,161         | 2,902,110         | 358,059       | 17,764       | 8,635      | 485,070         |
| Nanopore | 50×  | 2,598,083         | 2,914,706         | 360,142       | 17,602       | 8,754      | 493,774         |
| Nanopore | 60×  | 2,579,823         | 2,929,398         | 361,119       | 17,474       | 8,834      | 499,496         |
| HiFi     | 10×  | 2,367,609         | 2,081,641         | 425,786       | –            | –          | –               |
| HiFi     | 20×  | 2,485,185         | 2,318,132         | 442,122       | –            | –          | –               |
| HiFi     | 30×  | 2,468,170         | 2,428,526         | 432,578       | –            | –          | –               |
| HiFi     | 40×  | 2,438,078         | 2,508,667         | 425,166       | –            | –          | –               |
| HiFi     | 50×  | 2,410,815         | 2,566,571         | 420,096       | –            | –          | –               |

**Supplementary Table 8: SNV-only phasing of HG002 nanopore R10.4.1 data by six configurations.** Scored against the v5.0q VCF (chr1–22, no benchmark BED applied). Values at 10–20× are means (s.d.) over ten down-sampling replicates; 30–60×, one replicate. Phased SNVs are given as a percentage of the 2,398,880 heterozygous SNVs of the benchmark. Rate ratio, switch error rate divided by that of LongPhase 2 in the same down-sampling replicate. Margin and Ralph were run at 10–20× only; GCphase was not included (Methods). Ralph’s phase blocks sum to 4.3–5.8 Gb, more than the length of the autosomes, so its block N50 is not comparable with that of the other tools.

| Tool                    | Cov. | Phased SNV (%) | Switch errors | Switch error rate (%) | Rate ratio to LongPhase 2 | Hamming (%) | Blocks     | N50 (Mb)    |
|-------------------------|------|----------------|---------------|-----------------------|---------------------------|-------------|------------|-------------|
| LongPhase 2             | 10×  | 77.83 (0.07)   | 1,518 (29)    | 0.082 (0.002)         | –                         | 5.33 (0.25) | 7,143 (30) | 0.84 (0.01) |
|                         | 12×  | 84.48 (0.06)   | 1,335 (54)    | 0.066 (0.003)         | –                         | 5.41 (0.38) | 5,917 (52) | 1.10 (0.02) |
|                         | 14×  | 87.87 (0.05)   | 1,137 (39)    | 0.054 (0.002)         | –                         | 4.88 (0.58) | 5,284 (55) | 1.29 (0.03) |
|                         | 16×  | 89.58 (0.03)   | 1,011 (27)    | 0.047 (0.001)         | –                         | 4.20 (0.26) | 4,875 (44) | 1.44 (0.03) |
|                         | 18×  | 90.50 (0.04)   | 918 (18)      | 0.042 (0.001)         | –                         | 3.64 (0.26) | 4,608 (58) | 1.53 (0.03) |
|                         | 20×  | 90.94 (0.02)   | 823 (43)      | 0.038 (0.002)         | –                         | 3.21 (0.34) | 4,406 (46) | 1.61 (0.02) |
|                         | 30×  | 91.50          | 680           | 0.031                 | –                         | 1.82        | 3,793      | 1.92        |
|                         | 40×  | 91.54          | 602           | 0.027                 | –                         | 1.71        | 3,411      | 2.18        |
|                         | 50×  | 91.46          | 588           | 0.027                 | –                         | 1.48        | 3,198      | 2.51        |
|                         | 60×  | 91.35          | 496           | 0.023                 | –                         | 1.53        | 3,018      | 2.86        |
| LongPhase 2 without GNN | 10×  | 78.10 (0.07)   | 2,259 (39)    | 0.121 (0.002)         | 1.48 (0.02)               | 7.04 (0.29) | 6,056 (46) | 0.92 (0.01) |
|                         | 12×  | 84.71 (0.06)   | 1,995 (60)    | 0.098 (0.003)         | 1.49 (0.04)               | 7.21 (0.42) | 4,878 (39) | 1.21 (0.02) |
|                         | 14×  | 88.07 (0.05)   | 1,706 (50)    | 0.081 (0.002)         | 1.50 (0.04)               | 6.43 (0.55) | 4,314 (52) | 1.40 (0.02) |
|                         | 16×  | 89.75 (0.03)   | 1,514 (40)    | 0.070 (0.002)         | 1.50 (0.03)               | 5.51 (0.21) | 3,964 (43) | 1.54 (0.03) |
|                         | 18×  | 90.65 (0.03)   | 1,374 (36)    | 0.063 (0.002)         | 1.50 (0.04)               | 4.75 (0.22) | 3,740 (34) | 1.62 (0.03) |
|                         | 20×  | 91.08 (0.02)   | 1,235 (45)    | 0.057 (0.002)         | 1.50 (0.04)               | 4.12 (0.38) | 3,564 (35) | 1.70 (0.02) |
|                         | 30×  | 91.61          | 1,004         | 0.046                 | 1.47                      | 2.40        | 3,007      | 1.97        |
|                         | 40×  | 91.64          | 878           | 0.040                 | 1.46                      | 2.37        | 2,624      | 2.29        |
|                         | 50×  | 91.55          | 883           | 0.040                 | 1.50                      | 1.86        | 2,389      | 2.57        |
|                         | 60×  | 91.43          | 743           | 0.034                 | 1.50                      | 1.75        | 2,225      | 2.91        |
| WhatsHap (SNVs only)    | 10×  | 78.74 (0.07)   | 4,383 (65)    | 0.233 (0.003)         | 2.85 (0.06)               | 7.60 (0.39) | 5,960 (49) | 0.95 (0.01) |
|                         | 12×  | 85.35 (0.05)   | 4,229 (126)   | 0.207 (0.006)         | 3.14 (0.20)               | 7.99 (0.40) | 4,897 (45) | 1.24 (0.03) |
|                         | 14×  | 88.71 (0.05)   | 3,848 (127)   | 0.181 (0.006)         | 3.35 (0.10)               | 7.01 (0.42) | 4,435 (47) | 1.41 (0.02) |
|                         | 16×  | 90.37 (0.04)   | 3,561 (134)   | 0.165 (0.006)         | 3.49 (0.10)               | 6.37 (0.33) | 4,169 (52) | 1.53 (0.02) |
|                         | 18×  | 91.27 (0.03)   | 3,291 (134)   | 0.151 (0.006)         | 3.56 (0.15)               | 5.23 (0.39) | 4,009 (34) | 1.60 (0.04) |
|                         | 20×  | 91.68 (0.02)   | 3,067 (147)   | 0.140 (0.007)         | 3.71 (0.31)               | 4.40 (0.35) | 3,904 (38) | 1.65 (0.02) |
|                         | 30×  | 92.26          | 2,524         | 0.114                 | 3.68                      | 2.56        | 3,499      | 1.88        |
|                         | 40×  | 92.36          | 2,139         | 0.097                 | 3.52                      | 3.05        | 3,208      | 2.15        |
|                         | 50×  | 92.37          | 2,010         | 0.091                 | 3.38                      | 2.40        | 3,034      | 2.47        |
|                         | 60×  | 92.34          | 1,785         | 0.081                 | 3.56                      | 2.75        | 2,912      | 2.72        |
| HapCUT2                 | 10×  | 78.78 (0.09)   | 5,018 (108)   | 0.266 (0.006)         | 3.27 (0.07)               | 7.77 (0.39) | 5,878 (55) | 0.95 (0.01) |
|                         | 12×  | 85.41 (0.05)   | 4,948 (150)   | 0.242 (0.007)         | 3.67 (0.23)               | 8.06 (0.32) | 4,837 (41) | 1.25 (0.02) |
|                         | 14×  | 88.76 (0.07)   | 4,652 (104)   | 0.219 (0.005)         | 4.05 (0.12)               | 7.10 (0.52) | 4,382 (48) | 1.44 (0.03) |
|                         | 16×  | 90.44 (0.04)   | 4,344 (96)    | 0.201 (0.004)         | 4.26 (0.15)               | 6.13 (0.41) | 4,122 (50) | 1.54 (0.02) |

| Tool   | Cov. | Phased<br>SNV (%) | Switch<br>errors | Switch error<br>rate (%) | Rate ratio to<br>LongPhase 2 | Hamming<br>(%) | Blocks       | N50<br>(Mb)   |
|--------|------|-------------------|------------------|--------------------------|------------------------------|----------------|--------------|---------------|
|        | 18×  | 91.32 (0.05)      | 4,068 (121)      | 0.186 (0.006)            | 4.39 (0.12)                  | 5.32 (0.29)    | 3,969 (39)   | 1.61 (0.04)   |
|        | 20×  | 91.70 (0.03)      | 3,825 (140)      | 0.174 (0.006)            | 4.62 (0.34)                  | 4.27 (0.35)    | 3,872 (42)   | 1.66 (0.02)   |
|        | 30×  | 92.27             | 3,280            | 0.148                    | 4.78                         | 2.61           | 3,482        | 1.89          |
|        | 40×  | 92.37             | 2,916            | 0.132                    | 4.80                         | 2.90           | 3,203        | 2.16          |
|        | 50×  | 92.38             | 2,568            | 0.116                    | 4.32                         | 2.20           | 3,034        | 2.46          |
|        | 60×  | 92.34             | 2,309            | 0.104                    | 4.60                         | 2.11           | 2,911        | 2.72          |
| Margin | 10×  | 78.53 (0.07)      | 1,286 (31)       | 0.069 (0.002)            | 0.85 (0.03)                  | 0.47 (0.04)    | 18,177 (100) | 0.26 (0.00)   |
|        | 12×  | 85.13 (0.06)      | 1,347 (32)       | 0.066 (0.002)            | 1.01 (0.05)                  | 0.62 (0.10)    | 13,654 (88)  | 0.40 (0.01)   |
|        | 14×  | 88.47 (0.05)      | 1,331 (46)       | 0.063 (0.002)            | 1.17 (0.04)                  | 0.72 (0.05)    | 11,181 (69)  | 0.52 (0.01)   |
|        | 16×  | 90.14 (0.03)      | 1,302 (49)       | 0.061 (0.002)            | 1.29 (0.05)                  | 0.79 (0.11)    | 9,738 (58)   | 0.62 (0.01)   |
|        | 18×  | 91.04 (0.03)      | 1,281 (41)       | 0.059 (0.002)            | 1.39 (0.05)                  | 0.87 (0.10)    | 8,818 (49)   | 0.71 (0.00)   |
|        | 20×  | 91.47 (0.02)      | 1,254 (60)       | 0.057 (0.003)            | 1.52 (0.12)                  | 0.93 (0.11)    | 8,129 (46)   | 0.78 (0.01)   |
| Ralphi | 10×  | 78.52 (0.08)      | 3,784 (104)      | 0.201 (0.006)            | 2.47 (0.07)                  | 7.48 (0.27)    | 5,692 (49)   | 5.31 (7.10)   |
|        | 12×  | 85.14 (0.06)      | 3,694 (164)      | 0.181 (0.008)            | 2.75 (0.19)                  | 8.32 (0.53)    | 4,536 (34)   | 7.97 (8.42)   |
|        | 14×  | 88.50 (0.06)      | 3,452 (102)      | 0.163 (0.005)            | 3.02 (0.11)                  | 7.71 (0.63)    | 4,000 (58)   | 25.31 (16.84) |
|        | 16×  | 90.17 (0.04)      | 3,176 (104)      | 0.147 (0.005)            | 3.12 (0.12)                  | 7.60 (0.52)    | 3,681 (36)   | 30.67 (12.45) |
|        | 18×  | 91.07 (0.03)      | 2,979 (163)      | 0.137 (0.007)            | 3.23 (0.19)                  | 7.14 (0.51)    | 3,463 (30)   | 39.14 (14.83) |
|        | 20×  | 91.50 (0.02)      | 2,874 (96)       | 0.131 (0.004)            | 3.48 (0.25)                  | 6.56 (0.53)    | 3,314 (37)   | 47.11 (6.72)  |

**Supplementary Table 9: The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark.** Switch errors on SNVs; Hamming distance over SNVs and, where they are phased, indels. All LongPhase 2 runs with GNN correction, phasing SNVs alone or with indels (+indel) or indels and SVs (+indel+SV). Values at 10–20× are means (s.d.) over ten replicates. Phased indels as a percentage of benchmark heterozygous indels (v5.0q).

| Configuration            | Cov. | Switch errors<br>v4.2.1 | Switch errors<br>v5.0q | Hamming<br>v4.2.1 (%) | Hamming<br>v5.0q (%) | Phased<br>indels (%) |
|--------------------------|------|-------------------------|------------------------|-----------------------|----------------------|----------------------|
| LongPhase 2, SNVs        | 10×  | 1,820 (26)              | 1,518 (29)             | 5.30 (0.24)           | 5.33 (0.25)          | –                    |
|                          | 20×  | 1,656 (31)              | 823 (43)               | 3.12 (0.36)           | 3.21 (0.34)          | –                    |
|                          | 30×  | 1,635                   | 680                    | 1.80                  | 1.82                 | –                    |
|                          | 40×  | 1,663                   | 602                    | 1.64                  | 1.71                 | –                    |
|                          | 50×  | 1,669                   | 588                    | 1.43                  | 1.48                 | –                    |
|                          | 60×  | 1,676                   | 496                    | 1.54                  | 1.53                 | –                    |
| LongPhase 2, +indels     | 10×  | 1,972 (24)              | 1,704 (32)             | 6.80 (0.44)           | 6.85 (0.44)          | 30.42 (0.05)         |
|                          | 20×  | 1,807 (30)              | 996 (35)               | 5.75 (0.42)           | 5.78 (0.40)          | 41.03 (0.03)         |
|                          | 30×  | 1,777                   | 822                    | 4.72                  | 4.86                 | 43.38                |
|                          | 40×  | 1,758                   | 724                    | 4.22                  | 4.37                 | 44.39                |
|                          | 50×  | 1,752                   | 664                    | 3.13                  | 3.26                 | 45.02                |
|                          | 60×  | 1,765                   | 591                    | 3.53                  | 3.46                 | 45.35                |
| LongPhase 2, +indels+SVs | 10×  | 1,977 (24)              | 1,711 (30)             | 6.81 (0.41)           | 6.85 (0.41)          | 30.42 (0.05)         |
|                          | 20×  | 1,806 (33)              | 1,001 (33)             | 6.04 (0.51)           | 6.06 (0.50)          | 41.02 (0.03)         |
|                          | 30×  | 1,778                   | 838                    | 4.98                  | 5.07                 | 43.38                |
|                          | 40×  | 1,767                   | 739                    | 4.20                  | 4.38                 | 44.39                |
|                          | 50×  | 1,760                   | 673                    | 3.29                  | 3.39                 | 45.01                |
|                          | 60×  | 1,772                   | 592                    | 3.50                  | 3.46                 | 45.34                |
| WhatsHap, SNVs only      | 10×  | 2,773 (47)              | 4,383 (65)             | 7.44 (0.37)           | 7.60 (0.39)          | –                    |
|                          | 20×  | 1,972 (29)              | 3,067 (147)            | 4.21 (0.38)           | 4.40 (0.35)          | –                    |
|                          | 30×  | 1,850                   | 2,524                  | 2.45                  | 2.56                 | –                    |
|                          | 40×  | 1,810                   | 2,139                  | 2.92                  | 3.05                 | –                    |
|                          | 50×  | 1,811                   | 2,010                  | 2.23                  | 2.40                 | –                    |
|                          | 60×  | 1,801                   | 1,785                  | 2.61                  | 2.75                 | –                    |
| WhatsHap, SNVs + indels  | 10×  | 3,538 (48)              | 5,174 (78)             | 12.00 (0.34)          | 12.16 (0.36)         | 32.86 (0.03)         |
|                          | 20×  | 2,580 (32)              | 3,683 (122)            | 11.34 (0.59)          | 11.45 (0.57)         | 44.31 (0.04)         |
|                          | 30×  | 2,345                   | 3,027                  | 9.67                  | 9.71                 | 47.14                |
|                          | 40×  | 2,221                   | 2,572                  | 10.43                 | 10.40                | 48.52                |
|                          | 50×  | 2,171                   | 2,346                  | 8.32                  | 8.46                 | 49.36                |
|                          | 60×  | 2,156                   | 2,139                  | 9.20                  | 9.29                 | 49.84                |
| HapCUT2                  | 10×  | 3,050 (33)              | 5,018 (108)            | 7.59 (0.38)           | 7.77 (0.39)          | –                    |
|                          | 20×  | 2,346 (40)              | 3,825 (140)            | 4.09 (0.35)           | 4.27 (0.35)          | –                    |
|                          | 30×  | 2,202                   | 3,280                  | 2.46                  | 2.61                 | –                    |
|                          | 40×  | 2,171                   | 2,916                  | 2.72                  | 2.90                 | –                    |
|                          | 50×  | 2,133                   | 2,568                  | 2.03                  | 2.20                 | –                    |
|                          | 60×  | 2,088                   | 2,309                  | 2.02                  | 2.11                 | –                    |

**Supplementary Table 10: Co-phasing configurations of LongPhase 2, without and with GNN correction.** Scored against v5.0q: switch errors on SNVs, Hamming distance over SNVs and, where they are phased, indels; phased indels as a percentage of benchmark heterozygous indels. Means (s.d.) over ten replicates at 10 and 20×; one replicate at 30 and 60×. Every coverage is in Supplementary Data 1.

| Classes    | GNN | Cov. | Phased<br>indel (%) | Switch<br>errors | Switch error<br>rate (%) | Hamming<br>(%) | N50<br>(Mb) |
|------------|-----|------|---------------------|------------------|--------------------------|----------------|-------------|
| SNV        | no  | 10×  | –                   | 2,259 (39)       | 0.121 (0.002)            | 7.04 (0.29)    | 0.92 (0.01) |
|            |     | 20×  | –                   | 1,235 (45)       | 0.057 (0.002)            | 4.12 (0.38)    | 1.70 (0.02) |
|            |     | 30×  | –                   | 1,004            | 0.046                    | 2.40           | 1.97        |
|            |     | 60×  | –                   | 743              | 0.034                    | 1.75           | 2.91        |
|            | yes | 10×  | –                   | 1,518 (29)       | 0.082 (0.002)            | 5.33 (0.25)    | 0.84 (0.01) |
|            |     | 20×  | –                   | 823 (43)         | 0.038 (0.002)            | 3.21 (0.34)    | 1.61 (0.02) |
|            |     | 30×  | –                   | 680              | 0.031                    | 1.82           | 1.92        |
|            |     | 60×  | –                   | 496              | 0.023                    | 1.53           | 2.86        |
| +SV        | no  | 10×  | –                   | 2,292 (47)       | 0.123 (0.002)            | 7.23 (0.24)    | 0.93 (0.01) |
|            |     | 20×  | –                   | 1,262 (44)       | 0.058 (0.002)            | 4.44 (0.35)    | 1.72 (0.01) |
|            |     | 30×  | –                   | 1,031            | 0.047                    | 2.64           | 1.98        |
|            |     | 60×  | –                   | 745              | 0.034                    | 2.09           | 2.93        |
|            | yes | 10×  | –                   | 1,541 (34)       | 0.083 (0.002)            | 5.45 (0.24)    | 0.85 (0.01) |
|            |     | 20×  | –                   | 842 (44)         | 0.039 (0.002)            | 3.52 (0.32)    | 1.63 (0.02) |
|            |     | 30×  | –                   | 706              | 0.032                    | 1.96           | 1.93        |
|            |     | 60×  | –                   | 498              | 0.023                    | 1.68           | 2.90        |
| +5mC       | no  | 10×  | –                   | 2,247 (45)       | 0.120 (0.002)            | 7.13 (0.33)    | 0.94 (0.01) |
|            |     | 20×  | –                   | 1,228 (41)       | 0.056 (0.002)            | 4.38 (0.52)    | 1.84 (0.03) |
|            |     | 30×  | –                   | 1,015            | 0.046                    | 2.53           | 2.12        |
|            |     | 60×  | –                   | 736              | 0.034                    | 1.82           | 3.14        |
|            | yes | 10×  | –                   | 1,541 (32)       | 0.083 (0.002)            | 5.46 (0.27)    | 0.86 (0.01) |
|            |     | 20×  | –                   | 824 (40)         | 0.038 (0.002)            | 3.38 (0.35)    | 1.75 (0.02) |
|            |     | 30×  | –                   | 692              | 0.032                    | 2.01           | 2.07        |
|            |     | 60×  | –                   | 497              | 0.023                    | 1.60           | 3.09        |
| +indel     | no  | 10×  | 30.79 (0.04)        | 2,913 (52)       | 0.156 (0.003)            | 10.87 (0.39)   | 1.18 (0.02) |
|            |     | 20×  | 41.25 (0.03)        | 1,646 (47)       | 0.075 (0.002)            | 9.50 (0.38)    | 2.45 (0.05) |
|            |     | 30×  | 43.55               | 1,316            | 0.060                    | 8.02           | 2.96        |
|            |     | 60×  | 45.48               | 953              | 0.043                    | 6.38           | 4.35        |
|            | yes | 10×  | 30.42 (0.05)        | 1,704 (32)       | 0.092 (0.002)            | 6.85 (0.44)    | 0.97 (0.02) |
|            |     | 20×  | 41.03 (0.03)        | 996 (35)         | 0.046 (0.002)            | 5.78 (0.40)    | 2.01 (0.06) |
|            |     | 30×  | 43.38               | 822              | 0.037                    | 4.86           | 2.44        |
|            |     | 60×  | 45.35               | 591              | 0.027                    | 3.46           | 3.69        |
| +indel+SV  | no  | 10×  | 30.78 (0.04)        | 2,934 (51)       | 0.157 (0.003)            | 11.00 (0.39)   | 1.19 (0.02) |
|            |     | 20×  | 41.24 (0.03)        | 1,657 (51)       | 0.076 (0.002)            | 9.76 (0.43)    | 2.48 (0.04) |
|            |     | 30×  | 43.54               | 1,323            | 0.060                    | 8.45           | 2.99        |
|            |     | 60×  | 45.48               | 952              | 0.043                    | 6.51           | 4.37        |
|            | yes | 10×  | 30.42 (0.05)        | 1,711 (30)       | 0.092 (0.002)            | 6.85 (0.41)    | 0.98 (0.02) |
|            |     | 20×  | 41.02 (0.03)        | 1,001 (33)       | 0.046 (0.002)            | 6.06 (0.50)    | 2.04 (0.05) |
|            |     | 30×  | 43.38               | 838              | 0.038                    | 5.07           | 2.42        |
|            |     | 60×  | 45.34               | 592              | 0.027                    | 3.46           | 3.70        |
| +indel+5mC | no  | 10×  | 30.79 (0.05)        | 2,877 (48)       | 0.154 (0.003)            | 10.98 (0.37)   | 1.21 (0.02) |
|            |     | 20×  | 41.26 (0.03)        | 1,596 (46)       | 0.073 (0.002)            | 9.70 (0.55)    | 2.70 (0.08) |
|            |     | 30×  | 43.55               | 1,317            | 0.060                    | 8.01           | 3.19        |
|            |     | 60×  | 45.49               | 937              | 0.043                    | 7.33           | 5.03        |
|            | yes | 10×  | 30.48 (0.05)        | 1,724 (27)       | 0.093 (0.001)            | 7.04 (0.45)    | 0.99 (0.02) |
|            |     | 20×  | 41.08 (0.03)        | 972 (31)         | 0.045 (0.001)            | 5.99 (0.60)    | 2.21 (0.06) |
|            |     | 30×  | 43.41               | 842              | 0.038                    | 5.27           | 2.69        |
|            |     | 60×  | 45.39               | 593              | 0.027                    | 3.60           | 4.10        |
| all four   | no  | 10×  | 30.79 (0.05)        | 2,897 (49)       | 0.155 (0.003)            | 11.08 (0.43)   | 1.21 (0.02) |
|            |     | 20×  | 41.26 (0.03)        | 1,610 (52)       | 0.074 (0.002)            | 9.88 (0.62)    | 2.74 (0.07) |
|            |     | 30×  | 43.54               | 1,316            | 0.060                    | 8.33           | 3.19        |
|            |     | 60×  | 45.48               | 932              | 0.043                    | 7.23           | 5.10        |
|            | yes | 10×  | 30.47 (0.05)        | 1,726 (30)       | 0.093 (0.002)            | 7.06 (0.45)    | 1.00 (0.02) |
|            |     | 20×  | 41.08 (0.03)        | 981 (34)         | 0.045 (0.002)            | 6.13 (0.57)    | 2.24 (0.06) |
|            |     | 30×  | 43.41               | 863              | 0.039                    | 5.50           | 2.71        |
|            |     | 60×  | 45.38               | 585              | 0.027                    | 3.34           | 4.10        |

**Supplementary Table 11: SNV-only phasing of HG002 PacBio HiFi (Revio) data.** One replicate per coverage, scored against v5.0q. The network was trained on nanopore data only and applied without retraining. Variants were called separately at every coverage (Methods).

| Tool                    | Cov. | Phased SNV (%) | Switch errors | Switch error rate (%) | Hamming (%) | Blocks | N50 (kb) |
|-------------------------|------|----------------|---------------|-----------------------|-------------|--------|----------|
| LongPhase 2             | 10×  | 87.02          | 1,070         | 0.052                 | 0.78        | 15,916 | 341      |
|                         | 20×  | 91.73          | 945           | 0.043                 | 1.15        | 12,190 | 489      |
|                         | 30×  | 91.92          | 885           | 0.040                 | 0.82        | 11,164 | 541      |
|                         | 40×  | 91.89          | 869           | 0.040                 | 0.78        | 10,557 | 573      |
|                         | 50×  | 91.58          | 740           | 0.034                 | 0.89        | 10,215 | 584      |
| LongPhase 2 without GNN | 10×  | 87.12          | 1,328         | 0.064                 | 0.96        | 15,708 | 346      |
|                         | 20×  | 91.84          | 1,268         | 0.058                 | 1.28        | 11,980 | 492      |
|                         | 30×  | 92.02          | 1,137         | 0.052                 | 1.02        | 10,926 | 545      |
|                         | 40×  | 91.98          | 1,153         | 0.052                 | 0.98        | 10,347 | 580      |
|                         | 50×  | 91.66          | 988           | 0.045                 | 1.10        | 10,010 | 588      |
| WhatsHap (SNVs only)    | 10×  | 87.43          | 2,655         | 0.128                 | 1.03        | 15,804 | 340      |
|                         | 20×  | 92.20          | 2,504         | 0.114                 | 1.21        | 12,090 | 478      |
|                         | 30×  | 92.40          | 2,331         | 0.106                 | 0.92        | 11,082 | 525      |
|                         | 40×  | 92.37          | 1,954         | 0.089                 | 0.87        | 10,512 | 561      |
|                         | 50×  | 92.05          | 1,687         | 0.077                 | 0.68        | 10,186 | 577      |

**Supplementary Table 13: Location of SNV switch errors by truth set and benchmark region.** **a**, switch errors by truth set and region; **b**, by genomic stratum within the v5.0q benchmark regions; **c**, switch errors shared by the three tools. SNV-only phasing of HG002 nanopore R10.4.1 data; LongPhase 2 with GNN correction. A variant pair is assigned to a region when both of its variants lie in it, so region counts do not sum to the total. *Shared*, inside both sets of benchmark regions; *v5.0q only*, inside the v5.0q regions only; *outside*, outside both. Means over ten replicates at 10 and 20×; one replicate at 30 and 60×. Rates are switch errors per assessed SNV pair. Per-run values for every coverage, region and stratum are in Supplementary Data 1.

| Tool        | Cov. | v4.2.1 |        | v5.0q |              |                       |        |            |                   |
|-------------|------|--------|--------|-------|--------------|-----------------------|--------|------------|-------------------|
|             |      | All    | Shared | All   | Inside v5.0q | Rate inside v5.0q (%) | Shared | v5.0q only | Outside (rate, %) |
| LongPhase 2 | 10×  | 1,820  | 1,445  | 1,518 | 938          | 0.0527                | 805    | 41         | 438 (1.19)        |
|             | 20×  | 1,656  | 1,169  | 823   | 271          | 0.0130                | 196    | 30         | 446 (0.95)        |
|             | 30×  | 1,635  | 1,127  | 680   | 117          | 0.0056                | 62     | 25         | 454 (0.99)        |
|             | 60×  | 1,676  | 1,178  | 496   | 66           | 0.0032                | 33     | 15         | 348 (0.81)        |
| WhatsHap    | 10×  | 2,773  | 2,235  | 4,383 | 1,992        | 0.1112                | 1,483  | 266        | 2,030 (4.63)      |
|             | 20×  | 1,972  | 1,372  | 3,067 | 723          | 0.0347                | 340    | 240        | 2,065 (3.75)      |
|             | 30×  | 1,850  | 1,238  | 2,524 | 444          | 0.0211                | 134    | 205        | 1,829 (3.34)      |
|             | 60×  | 1,801  | 1,245  | 1,785 | 235          | 0.0112                | 63     | 88         | 1,337 (2.61)      |
| HapCUT2     | 10×  | 3,050  | 2,432  | 5,018 | 2,445        | 0.1364                | 1,686  | 322        | 2,175 (4.93)      |
|             | 20×  | 2,346  | 1,633  | 3,825 | 1,264        | 0.0606                | 613    | 315        | 2,262 (4.10)      |
|             | 30×  | 2,202  | 1,492  | 3,280 | 937          | 0.0446                | 393    | 268        | 2,053 (3.74)      |
|             | 60×  | 2,088  | 1,435  | 2,309 | 676          | 0.0321                | 273    | 148        | 1,430 (2.78)      |

**b** Within the v5.0q benchmark regions, by GIAB v3.6 stratum

| Tool        | Cov. | Segmental duplication | Low mappability or seg. dup. | Tandem repeat | Homopolymer | Satellite | Not difficult |
|-------------|------|-----------------------|------------------------------|---------------|-------------|-----------|---------------|
| LongPhase 2 | 10×  | 51                    | 55                           | 5             | 1           | 2         | 604           |
|             | 20×  | 37                    | 39                           | 3             | 0           | 2         | 135           |
|             | 30×  | 29                    | 29                           | 0             | 0           | 0         | 45            |
|             | 60×  | 18                    | 19                           | 0             | 0           | 0         | 20            |
| WhatsHap    | 10×  | 264                   | 279                          | 33            | 6           | 12        | 1,079         |
|             | 20×  | 232                   | 239                          | 29            | 5           | 10        | 240           |
|             | 30×  | 202                   | 205                          | 16            | 5           | 3         | 91            |
|             | 60×  | 80                    | 82                           | 15            | 0           | 0         | 38            |
| HapCUT2     | 10×  | 292                   | 308                          | 54            | 19          | 12        | 1,040         |
|             | 20×  | 275                   | 283                          | 45            | 21          | 10        | 244           |
|             | 30×  | 247                   | 254                          | 36            | 19          | 6         | 100           |
|             | 60×  | 120                   | 124                          | 25            | 21          | 3         | 40            |

**c** Switch errors at identical SNV pairs across tools, replicate 1

| Truth  | Cov. | LongPhase 2 | WhatsHap | HapCUT2 | All three | All three<br>(% of LongPhase 2) |
|--------|------|-------------|----------|---------|-----------|---------------------------------|
| v4.2.1 | 10×  | 1,829       | 2,775    | 3,041   | 1,145     | 63                              |
|        | 20×  | 1,680       | 1,922    | 2,292   | 1,405     | 84                              |
|        | 30×  | 1,635       | 1,850    | 2,202   | 1,474     | 90                              |
|        | 60×  | 1,676       | 1,801    | 2,088   | 1,556     | 93                              |
| v5.0q  | 10×  | 1,510       | 4,258    | 4,920   | 543       | 36                              |
|        | 20×  | 922         | 2,735    | 3,552   | 289       | 31                              |
|        | 30×  | 680         | 2,524    | 3,280   | 236       | 35                              |
|        | 60×  | 496         | 1,785    | 2,309   | 183       | 37                              |

**Supplementary Table 14: Phase accuracy of indels against v5.0q.** HG002 nanopore R10.4.1 data, co-phasing runs. Indel-pair switch errors are counted over consecutive variant pairs of an intersected block that include at least one indel; the indel Hamming distance uses the block orientation chosen on all variants. Truth VCF used in full; the last column restricts the indel-pair rate to the v5.0q benchmark regions. Means (s.d.) over ten replicates at 10 and 20×; one replicate at 30 and 60×. Per-run values are in Supplementary Data 1.

| Configuration             | Cov. | Phased<br>indels | Indel-pair<br>switch errors | Indel-pair<br>rate (%) | SNV+indel<br>rate (%) | Indel<br>Hamming (%) | Indel-pair rate,<br>inside v5.0q (%) |
|---------------------------|------|------------------|-----------------------------|------------------------|-----------------------|----------------------|--------------------------------------|
| LongPhase 2, +indel       | 10×  | 157,148 (247)    | 300 (23)                    | 0.101 (0.008)          | 0.090 (0.001)         | 6.78 (0.46)          | 0.080 (0.007)                        |
|                           | 20×  | 211,933 (164)    | 206 (10)                    | 0.052 (0.003)          | 0.045 (0.001)         | 5.72 (0.37)          | 0.032 (0.003)                        |
|                           | 30×  | 224,081          | 179                         | 0.043                  | 0.037                 | 4.89                 | 0.023                                |
|                           | 60×  | 234,254          | 130                         | 0.030                  | 0.027                 | 3.52                 | 0.015                                |
| same, no GNN              | 10×  | 159,017 (231)    | 495 (31)                    | 0.165 (0.010)          | 0.150 (0.003)         | 10.83 (0.41)         | 0.133 (0.009)                        |
|                           | 20×  | 213,062 (161)    | 305 (10)                    | 0.076 (0.002)          | 0.073 (0.002)         | 9.49 (0.34)          | 0.050 (0.003)                        |
|                           | 30×  | 224,926          | 236                         | 0.056                  | 0.058                 | 8.08                 | 0.031                                |
|                           | 60×  | 234,940          | 183                         | 0.042                  | 0.042                 | 6.40                 | 0.022                                |
| LongPhase 2, four classes | 10×  | 157,408 (252)    | 311 (21)                    | 0.104 (0.007)          | 0.091 (0.001)         | 6.98 (0.43)          | 0.084 (0.005)                        |
|                           | 20×  | 212,187 (161)    | 200 (11)                    | 0.050 (0.003)          | 0.044 (0.001)         | 6.08 (0.53)          | 0.030 (0.003)                        |
|                           | 30×  | 224,236          | 181                         | 0.043                  | 0.039                 | 5.59                 | 0.022                                |
|                           | 60×  | 234,414          | 138                         | 0.031                  | 0.027                 | 3.46                 | 0.015                                |
| WhatsHap, +indel          | 10×  | 169,742 (173)    | 2,685 (57)                  | 0.840 (0.018)          | 0.350 (0.005)         | 12.54 (0.32)         | 0.734 (0.014)                        |
|                           | 20×  | 228,884 (187)    | 3,379 (55)                  | 0.788 (0.013)          | 0.272 (0.006)         | 11.91 (0.62)         | 0.700 (0.013)                        |
|                           | 30×  | 243,513          | 3,764                       | 0.827                  | 0.262                 | 10.37                | 0.751                                |
|                           | 60×  | 257,426          | 4,377                       | 0.911                  | 0.253                 | 9.81                 | 0.858                                |

**Supplementary Table 15: SNV-only phasing on the chromosomes held out from GNN training (chr17, chr21 and chr22).** HG002 nanopore R10.4.1 data; the phased VCFs of the genome-wide analysis scored with `longphase compare -regions=chr17,chr21,chr22` against each truth VCF in full. Switch errors of LongPhase 2 as mean (s.d.); switch error rates in %. *Removed*, switch errors removed by GNN correction relative to LongPhase 2 without it; *cost*, phased SNVs withheld per removed switch error (the deployment criterion requires at most 10; Supplementary Method 4). Means over ten replicates at 10–20×; one replicate at 30–60×. Margin and Ralphi were run at 10–20× only.

| Truth  | Cov. | LongPhase 2   |          | Switch error rate (%) |          |         |        |        | GNN correction |      |
|--------|------|---------------|----------|-----------------------|----------|---------|--------|--------|----------------|------|
|        |      | Switch errors | Rate (%) | without GNN           | WhatsHap | HapCUT2 | Margin | Ralphi | Removed (%)    | Cost |
| v5.0q  | 10×  | 126 (16)      | 0.127    | 0.163                 | 0.360    | 0.416   | 0.099  | 0.347  | 22.0           | 10.4 |
|        | 12×  | 116 (16)      | 0.107    | 0.133                 | 0.339    | 0.390   | 0.095  | 0.322  | 20.2           | 11.5 |
|        | 14×  | 92 (12)       | 0.082    | 0.106                 | 0.288    | 0.334   | 0.089  | 0.282  | 23.1           | 10.6 |
|        | 16×  | 90 (10)       | 0.078    | 0.099                 | 0.258    | 0.316   | 0.083  | 0.248  | 21.3           | 10.0 |
|        | 18×  | 84 (8)        | 0.072    | 0.089                 | 0.243    | 0.303   | 0.081  | 0.249  | 19.5           | 10.1 |
|        | 20×  | 71 (16)       | 0.060    | 0.076                 | 0.233    | 0.297   | 0.076  | 0.243  | 20.9           | 9.7  |
|        | 30×  | 83            | 0.070    | 0.081                 | 0.196    | 0.212   | –      | –      | 13.5           | 12.2 |
|        | 40×  | 80            | 0.067    | 0.080                 | 0.179    | 0.187   | –      | –      | 15.8           | 7.3  |
|        | 50×  | 61            | 0.052    | 0.061                 | 0.152    | 0.182   | –      | –      | 15.3           | 11.9 |
|        | 60×  | 37            | 0.031    | 0.039                 | 0.144    | 0.154   | –      | –      | 19.6           | 15.1 |
| v4.2.1 | 10×  | 134 (8)       | 0.149    | 0.178                 | 0.215    | 0.225   | 0.080  | 0.195  | 16.4           | 9.3  |
|        | 12×  | 128 (8)       | 0.131    | 0.153                 | 0.180    | 0.193   | 0.081  | 0.172  | 14.4           | 9.1  |
|        | 14×  | 126 (6)       | 0.123    | 0.138                 | 0.166    | 0.176   | 0.085  | 0.155  | 11.0           | 9.9  |
|        | 16×  | 121 (7)       | 0.116    | 0.128                 | 0.146    | 0.159   | 0.086  | 0.142  | 9.6            | 9.3  |
|        | 18×  | 121 (5)       | 0.115    | 0.122                 | 0.138    | 0.151   | 0.092  | 0.137  | 5.8            | 12.4 |
|        | 20×  | 120 (6)       | 0.113    | 0.118                 | 0.134    | 0.147   | 0.093  | 0.132  | 4.2            | 12.3 |
|        | 30×  | 118           | 0.110    | 0.114                 | 0.128    | 0.140   | –      | –      | 3.3            | 9.2  |

| Truth | Cov. | LongPhase 2   |          | Switch error rate (%) |          |         |        |        | GNN correction |      |
|-------|------|---------------|----------|-----------------------|----------|---------|--------|--------|----------------|------|
|       |      | Switch errors | Rate (%) | without GNN           | WhatsHap | HapCUT2 | Margin | Ralphi | Removed (%)    | Cost |
|       | 40×  | 112           | 0.104    | 0.105                 | 0.118    | 0.128   | –      | –      | 0.9            | 61.0 |
|       | 50×  | 117           | 0.109    | 0.109                 | 0.117    | 0.125   | –      | –      | 0.0            | –    |
|       | 60×  | 117           | 0.109    | 0.110                 | 0.118    | 0.126   | –      | –      | 0.8            | 24.0 |

**Supplementary Data 1.** `supplementary.xlsx`: per-run metrics behind Figs. 2–6, Supplementary Figs. 10–16 and Supplementary Tables 7–11, 13 and 14, with one row per coverage and down-sampling replicate. Sheet `Variant_Calling`, input call sets; `SNV_Detail`, SNV-only phasing by every tool on nanopore and HiFi data; `co-phase_Detail`, the six co-phasing configurations of LongPhase 2 without (`longphase_v2.0.1`) and with (`longphase_v2.1`) GNN correction, and WhatsHap with indels; `SNV_switch_by_region`, SNV switch errors and assessed pairs of SNV-only runs per truth set, benchmark region and GIAB v3.6 stratum; `SNV_switch_coincidence`, switch errors at identical SNV pairs across tools (replicate 1); `Indel_accuracy`, indel and SNV switch errors, assessed pairs and Hamming distance of the co-phasing runs, without and within the v5.0q benchmark regions. Blank cells are runs that were not performed.

**Supplementary Table 12: SNVs unphased by GNN correction, classified by the T2T-HG002 v1.1 assembly.** Nanopore 60×, replicate 1, SNV-only phasing. “Aligned” means inside the dipcall BED in which both assembled haplotypes align 1:1 to GRCh38 (GRCh38\_HG2-T2TQ100-V1.1\_dipcall-z2k.dip.bed); presence of an assembly variant record at the same start coordinate in the matching dipcall VCF (alleles not compared). All chromosomes, including chrX and chrY, unless stated. The background is a random sample of 50,000 SNVs phased before GNN correction, including sites that GNN correction later unphased. Top, overall classes; bottom, share outside the 1:1 alignment per chromosome. v5.0q is derived from the same assembly, so this analysis extends coverage rather than providing an independent truth set.

| Class                                                      | SNVs            | Share (%) |
|------------------------------------------------------------|-----------------|-----------|
| not aligned 1:1 (unassessed)                               | 49,475          | 88.23     |
| aligned, no assembly variant record at the same coordinate | 3,421           | 6.10      |
| aligned, assembly variant record at the same coordinate    | 3,180           | 5.67      |
| total                                                      | 56,076          | 100.00    |
| aligned 1:1, unphased by GNN correction                    | 6,601 / 56,076  | 11.77     |
| aligned 1:1, phased before GNN correction (sample)         | 46,318 / 50,000 | 92.64     |
| autosomes only, unphased by GNN correction                 | 4,955 / 51,595  | 9.60      |
| autosomes only, phased before GNN correction (sample)      | 46,116 / 49,622 | 92.93     |
| inside the v5.0q benchmark BED                             | 549             | —         |
| of which benchmark variant                                 | 413             | 75.23     |

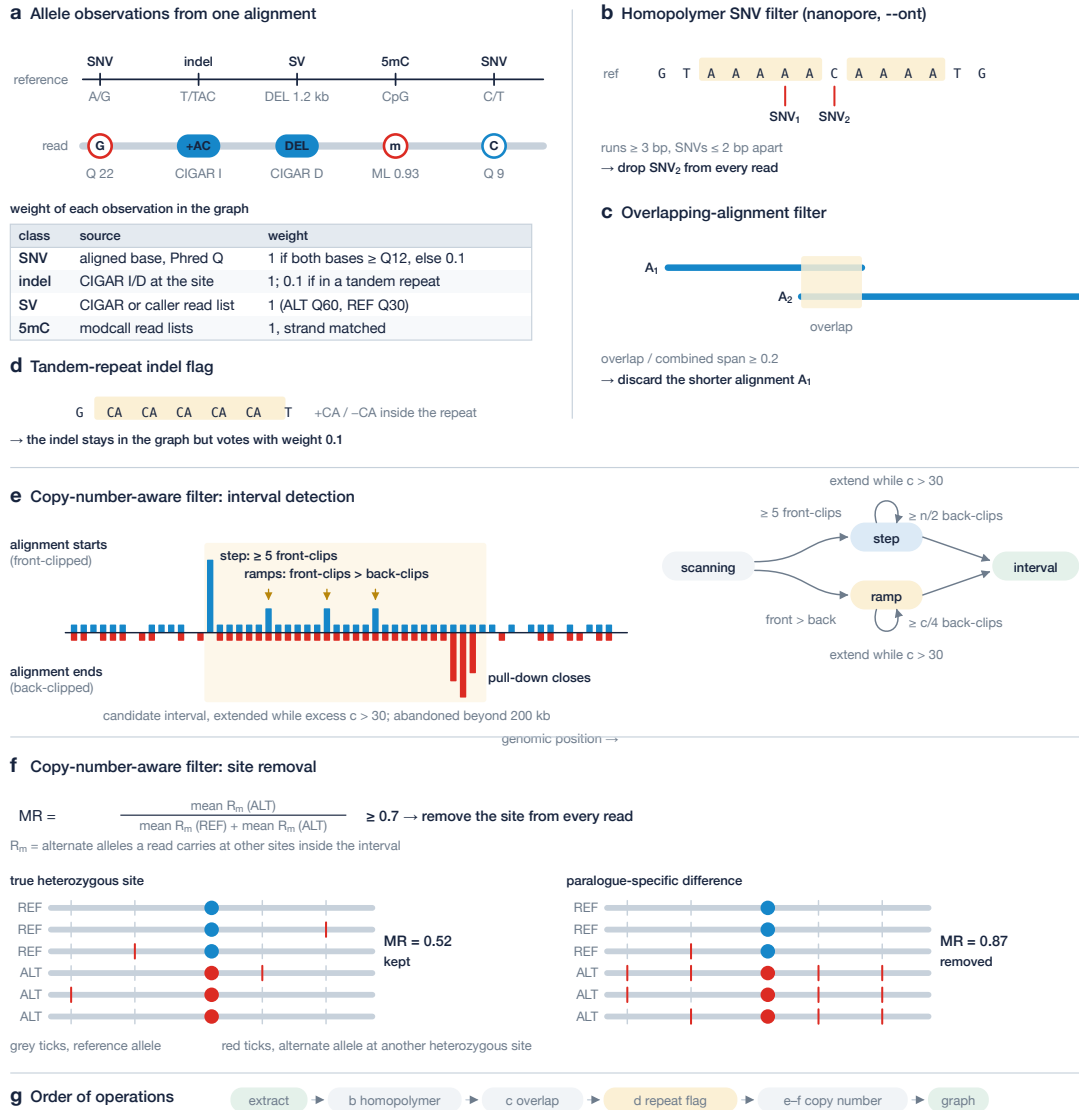
  

| Chromosome | Phased before GNN correction<br><i>n</i> | not aligned (%) | Unphased by GNN correction<br><i>n</i> | not aligned (%) | Ratio |
|------------|------------------------------------------|-----------------|----------------------------------------|-----------------|-------|
| chr1       | 3,787                                    | 7.05            | 3,877                                  | 88.26           | 12.5  |
| chr2       | 3,713                                    | 3.42            | 1,930                                  | 92.69           | 27.1  |
| chr3       | 3,467                                    | 4.99            | 2,410                                  | 80.12           | 16.1  |
| chr4       | 3,355                                    | 3.37            | 1,188                                  | 75.93           | 22.5  |
| chr5       | 3,184                                    | 2.73            | 1,249                                  | 61.89           | 22.7  |
| chr6       | 3,135                                    | 2.90            | 676                                    | 88.61           | 30.6  |
| chr7       | 2,949                                    | 5.49            | 3,060                                  | 77.65           | 14.1  |
| chr8       | 2,707                                    | 5.13            | 1,872                                  | 91.19           | 17.8  |
| chr9       | 2,342                                    | 12.34           | 1,689                                  | 92.30           | 7.5   |
| chr10      | 2,478                                    | 4.16            | 1,530                                  | 89.80           | 21.6  |
| chr11      | 2,639                                    | 13.19           | 10,385                                 | 99.06           | 7.5   |
| chr12      | 2,420                                    | 9.13            | 4,374                                  | 99.73           | 10.9  |
| chr13      | 1,801                                    | 6.11            | 2,268                                  | 96.69           | 15.8  |
| chr14      | 1,590                                    | 2.01            | 339                                    | 52.21           | 26.0  |
| chr15      | 1,482                                    | 8.70            | 1,096                                  | 78.01           | 9.0   |
| chr16      | 1,496                                    | 8.29            | 2,660                                  | 85.64           | 10.3  |
| chr17      | 1,512                                    | 11.11           | 1,373                                  | 67.81           | 6.1   |
| chr18      | 1,679                                    | 24.84           | 6,988                                  | 98.10           | 3.9   |
| chr19      | 1,130                                    | 0.09            | 138                                    | 42.75           | 475.0 |
| chr20      | 1,219                                    | 11.16           | 1,627                                  | 88.57           | 7.9   |
| chr21      | 781                                      | 20.23           | 545                                    | 88.99           | 4.4   |
| chr22      | 756                                      | 14.68           | 321                                    | 88.79           | 6.0   |
| chrX       | 263                                      | 44.11           | 4,006                                  | 68.50           | 1.6   |
| chrY       | 115                                      | 52.17           | 475                                    | 19.16           | 0.4   |

**Supplementary Table 16: Runtime and peak memory of SNV-only phasing.** HG002 nanopore R10.4.1 data, replicate 1 at 10× and the 60× data set, on the workstation described in Methods. Wall-clock time, CPU time (user plus system) and peak resident memory (GiB) from `/usr/bin/time`. LongPhase 2 includes GNN correction; in the 24-thread runs, phasing and GNN correction were run as separate commands and their times summed (Methods). HapCUT2 includes fragment extraction.

| Tool        | Threads | 10×        |           |        | 60×        |           |        |
|-------------|---------|------------|-----------|--------|------------|-----------|--------|
|             |         | Wall (min) | CPU (min) | Memory | Wall (min) | CPU (min) | Memory |
| LongPhase 2 | 1       | 8.1        | 8.3       | 6.2    | 18.0       | 18.8      | 7.4    |
| LongPhase 2 | 24      | 4.6        | 13.6      | 36.0   | 16.3       | 34.5      | 63.0   |
| WhatsHap    | 1       | 25.2       | 25.1      | 1.5    | 113.8      | 113.6     | 4.0    |
| HapCUT2     | 1       | 24.1       | 24.1      | 6.6    | 96.4       | 95.9      | 19.4   |

## Supplementary Figures

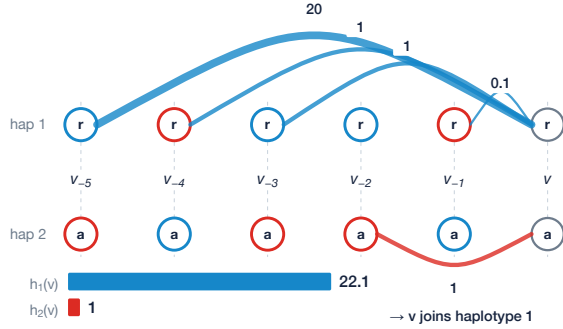


**Supplementary Fig. 1: Read-level allele observations and pre-graph filters.** **a**, Allele observations extracted from one alignment for each variant class, and the weight of each observation in the phasing graph (Supplementary Method 2). **b**, Homopolymer SNV filter (`--ont`): of two SNVs in homopolymers of length  $\geq 3$  that are  $\leq 2$  bp apart, the downstream SNV is removed. **c**, Overlapping-alignment filter for supplementary alignments of one read. **d**, Indels in tandem repeats remain in the graph and vote with weight 0.1. **e**, Clusters of clipped alignments define candidate intervals for the copy-number-aware filter (Supplementary Method 2). **f**, Within candidate intervals, heterozygous sites with  $MR \geq 0.7$  are removed from all reads; MR is the mean alternate-allele load of the reads carrying the alternate allele relative to the summed means of both read groups. **g**, Order in which the filters are applied.

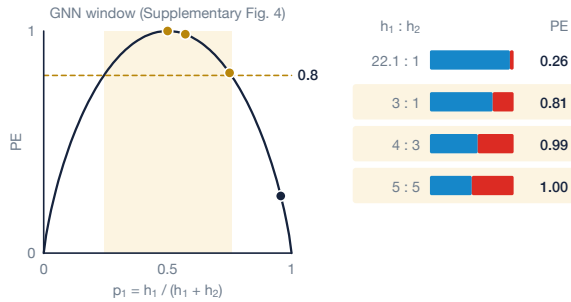
**a Allele-pair support between  $u$  and  $v$**



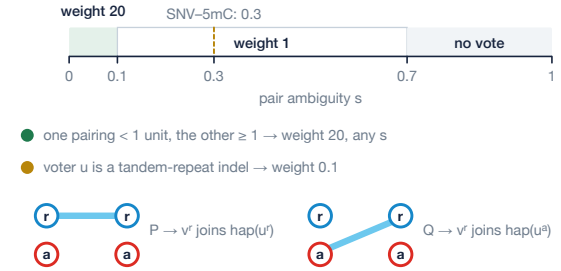
**c Weighted votes from up to  $k = 35$  upstream variants**



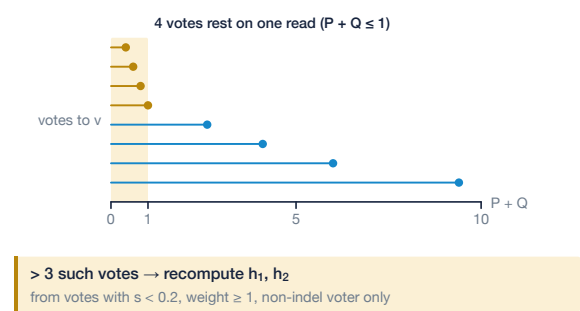
**e Phasing entropy  $PE = -\sum p_i \log_2 p_i$**



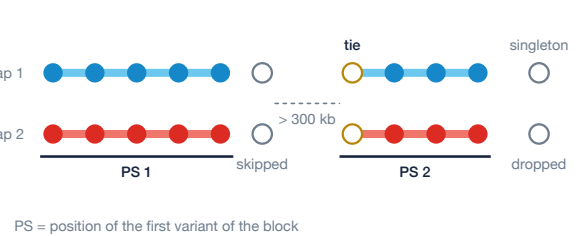
**b Vote rule for the pair ( $u, v$ )**



**d Single-read guard**

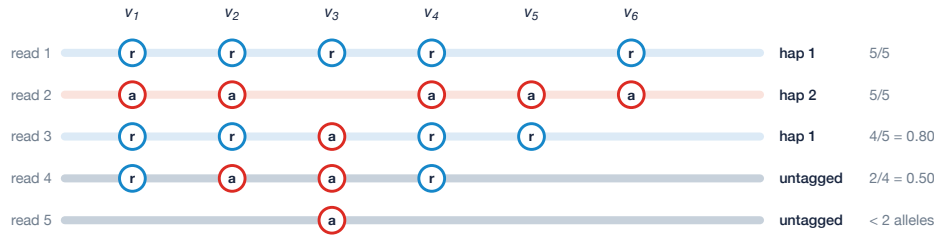


**f Block formation**



**Supplementary Fig. 2: Pairwise allele support, weighted voting and phasing entropy.** **a**, Reads spanning two variants  $u$  and  $v$  accumulate four weights;  $P$  and  $Q$  are the supports for the two possible pairings and  $s = \min(P, Q) / \max(P, Q)$  is their ambiguity. **b**, Vote weight as a function of  $s$ ; the rules are given in Methods and Supplementary Method 3. **c**, Variant  $v$  receives weighted votes from up to 35 upstream variants that share a read with it; line thickness encodes vote weight, and the totals  $h_1$  and  $h_2$  decide its haplotype. **d**, Single-read guard: when more than three votes rest on one read,  $h_1$  and  $h_2$  are recomputed from well-supported votes only. **e**, Phasing entropy (PE) of the vote totals as a function of the vote split (left) and for four examples (right); values  $\geq 0.8$  (shaded) open a window for the network (Supplementary Fig. 4). **f**, Block formation; the phase set (PS) is the position of the first variant of a block.

**a Step 1: read assignment to haplotypes**

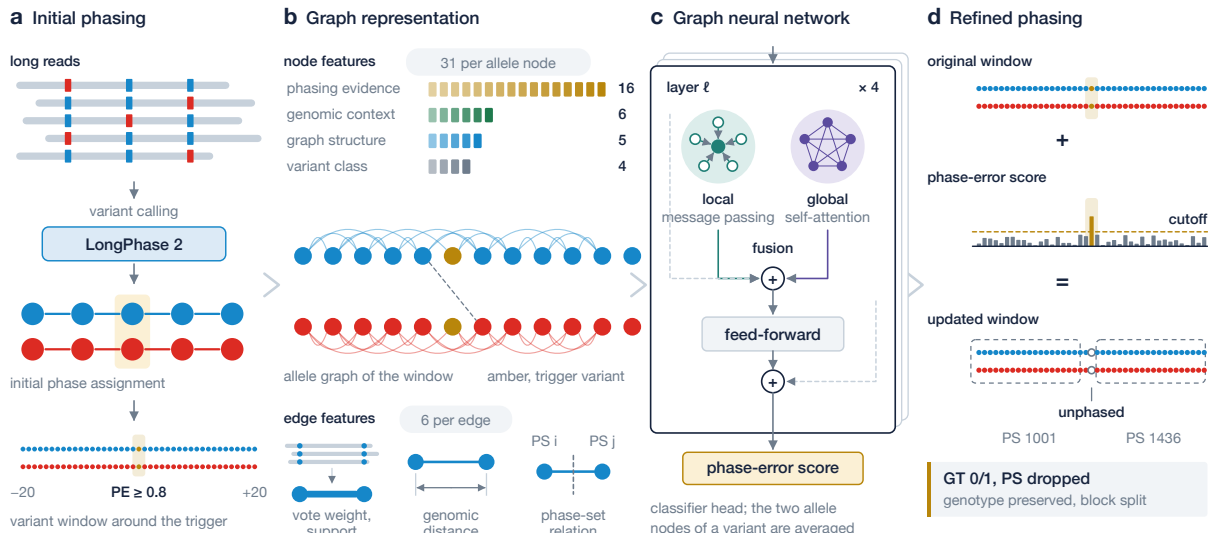


tagged if  $\max(n_1, n_2) / (n_1 + n_2) > 0.65$  and weighted evidence  $> 1$   
 allele weights: SNV 1, SV 1, indel 0.1, 5mC 0 (so  $\geq 2$  SNV/SV alleles are needed)

**b Step 2: phase re-derivation from tagged reads**

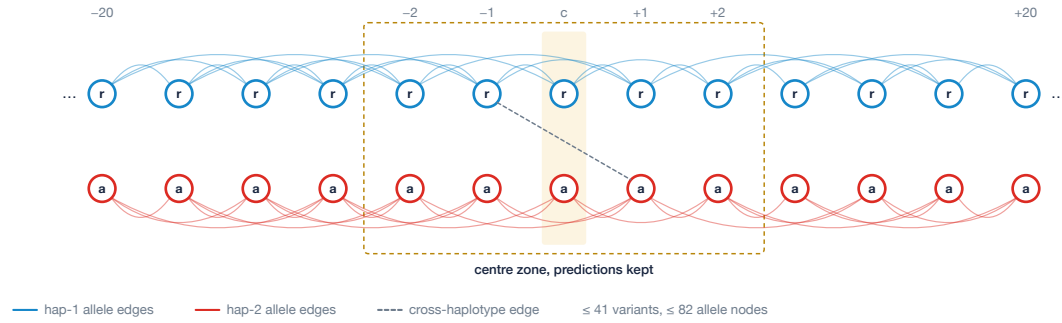


**Supplementary Fig. 3: Read-based correction.** **a**, Each read is assigned to a haplotype by a weighted majority over its phased alleles; reads with a majority  $\leq 0.65$  or fewer than two informative alleles remain untagged. **b**, The phase of each variant is re-derived from the tagged reads and retained only if the consistent fraction  $\rho$  exceeds 0.75; otherwise the variant is unphased.

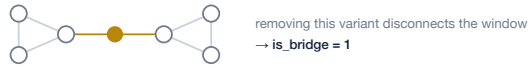


**Supplementary Fig. 4: Overview of the graph neural network correction module.** **a**, Every phased variant with a phasing entropy of at least 0.8 (amber) triggers a window of  $\pm 20$  variants. **b**, The window is represented as a graph of allele nodes with 31 node features and six edge features (Supplementary Tables 1 and 2); squares represent features, not their values. **c**, Each of four GPS layers combines a local GATv2 branch with a global self-attention branch; a classifier head assigns each allele node a phase-error score. **d**, Variants with a phase-error score of at least 0.30 are unphased without changing their genotypes, and the phase set is split where the remaining graph is disconnected. Details are shown in Supplementary Figs. 5 and 6.

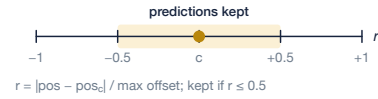
**a** Window around a trigger variant ( $PE \geq 0.8$ )



**b** Bridge-vertex feature



**c** Centre-zone mask

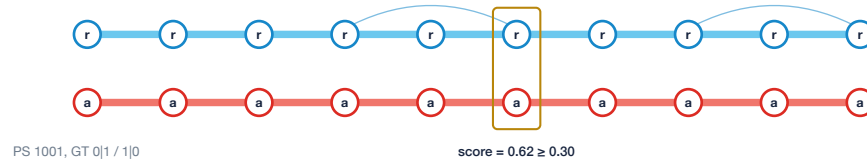


**d** Exported phasing graph (DOT)

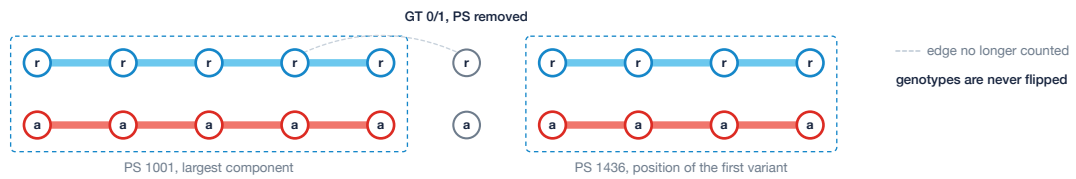
```
1001.1 -> 1543.1 [label=16.0] 1001.2 -> 1543.2 [label=1.1]
```

position.allele → position.allele, label = vote weight cast at the source

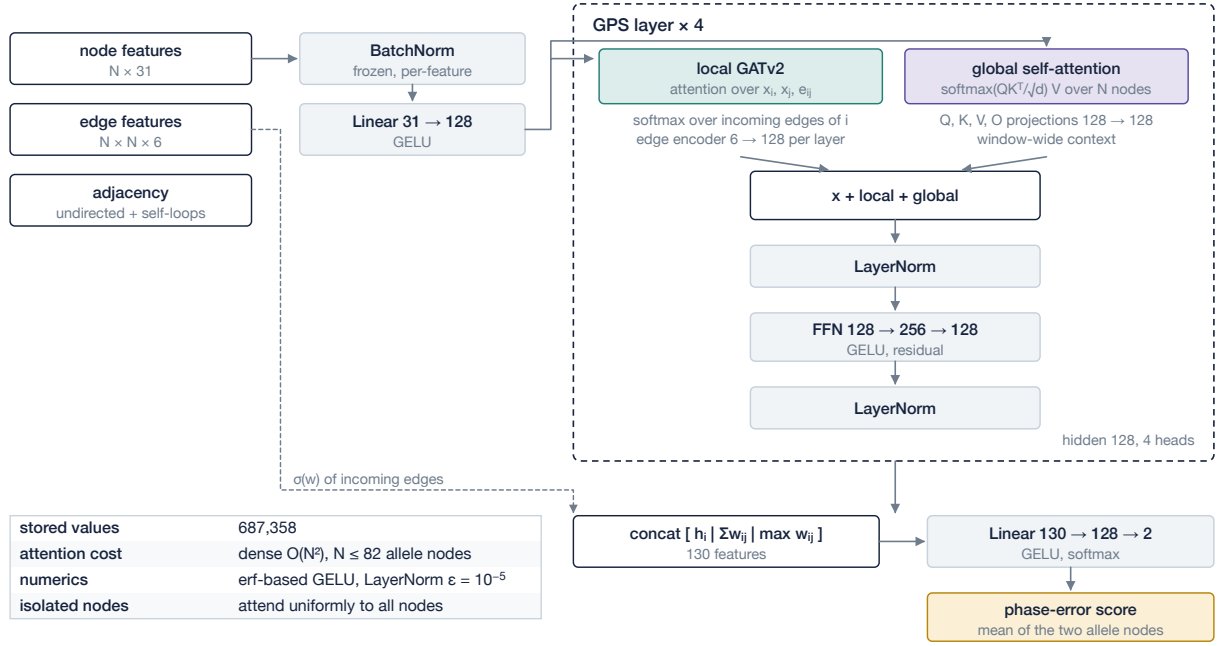
**e** Before correction: one phase set



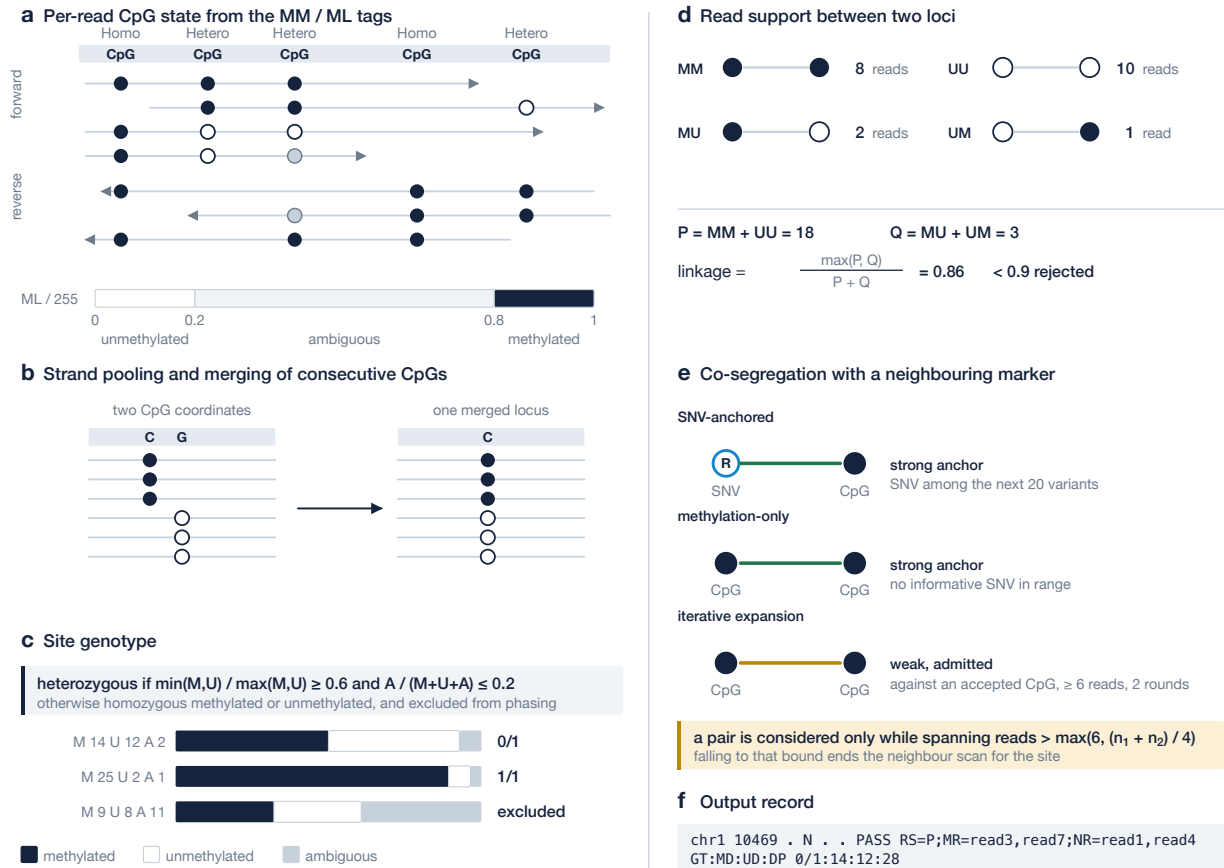
**f** After correction: variant unphased, phase set split



**Supplementary Fig. 5: Window construction and phase-set update for the graph neural network.** **a**, Window of  $\pm 20$  variants around a trigger with  $PE \geq 0.8$ ; arcs are phasing-graph edges and the dashed grey edge is a cross-haplotype link. **b**, Bridge-vertex feature: a variant whose removal disconnects the window graph. **c**, Only variants within half the largest distance from the trigger contribute to the training loss and to inference scores; the other variants supply graph context. **d**, The phasing graph in its DOT export format (`-dot`), with the vote weight at the source as the edge label. **e**, Before correction: a variant with a phase-error score  $\geq 0.30$  (amber) inside a phase set. **f**, After correction: the variant is unphased; the largest remaining component keeps the original PS and every other component receives the position of its first variant as its PS.

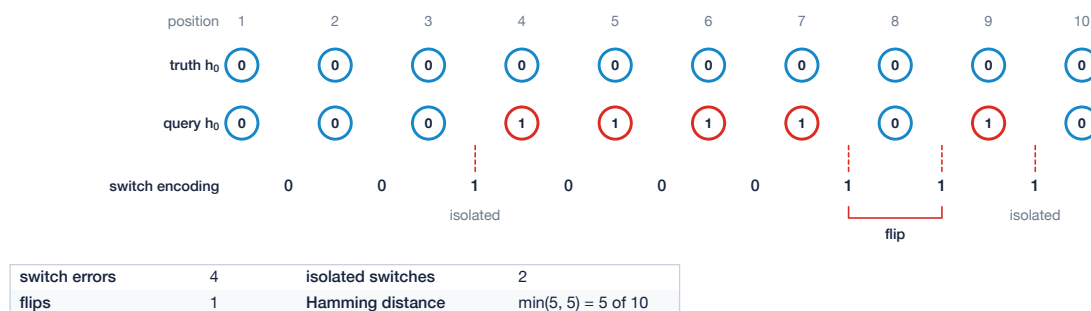


**Supplementary Fig. 6: Graph neural network architecture.** Node features are standardized and projected to 128 dimensions. Each of four GPS layers combines a local GATv2 branch over the window graph, with edge features entering through a per-layer edge encoder, and a global self-attention branch over all nodes; the branches are added to the residual stream and followed by layer normalization and a feed-forward block. The final embedding is concatenated with the sum and the maximum of the logistic-transformed weights  $\sigma(w) = 1/(1 + e^{-w})$  of the incoming edges, excluding the self-loop, and passed to a two-class classifier whose error-class output is the phase-error score; the scores of the two allele nodes of a variant are averaged.



**Supplementary Fig. 7: Allele-specific methylation calling with modcall.** **a**, Per-read CpG state from the MM/ML tags: methylated at  $ML/255 \geq 0.8$ , unmethylated at  $\leq 0.2$ , ambiguous in between. **b**, Strand-specific CpG observations are combined into loci for phasing: reverse-strand calls are mapped to the forward CpG coordinate and consecutive CpGs are merged. **c**, A CpG is genotyped as a candidate allele-specific site (0/1) when  $\min(M, U) / \max(M, U) \geq 0.6$  and  $A / (M + U + A) \leq 0.2$ , for methylated, unmethylated and ambiguous read counts  $M$ ,  $U$  and  $A$ ; other sites are excluded. **d**, Read support between two loci in the four state pairs; a pair is linked when  $\max(P, Q) / (P + Q) \geq 0.9$ , and the example shown is rejected. **e**, Linking against an SNV among the next 20 variants, against another candidate CpG where no SNV is in range, and expansion to weak candidates (Supplementary Method 1). **f**, Output record; the MR and NR read names attach the 5mC alleles to reads.

**a** Switch errors, flips and Hamming distance within one block



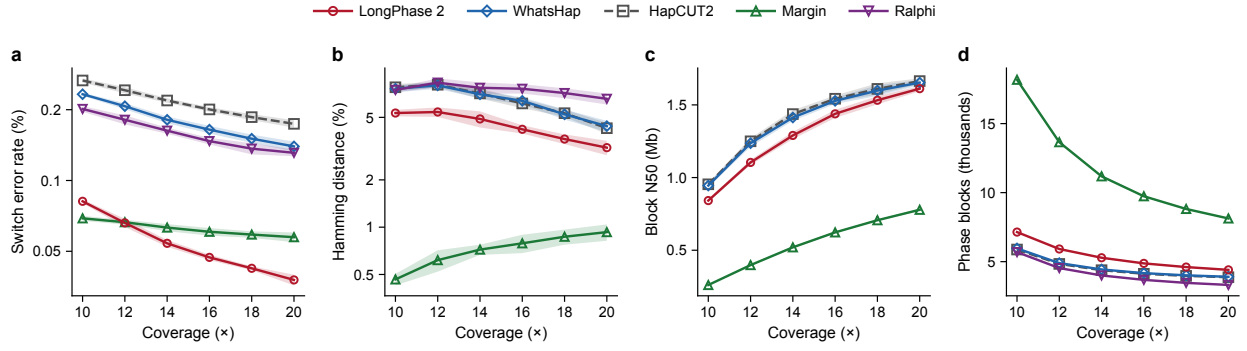
**b** Block statistics



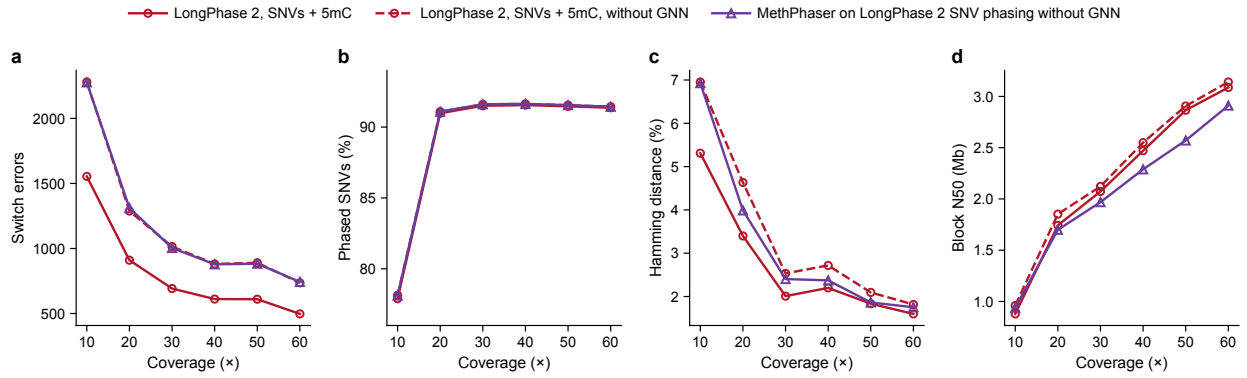
**Supplementary Fig. 8: Phasing evaluation metrics.** **a**, Within one intersected block, the query haplotype string is compared with the truth in switch encoding; the number of differing positions is the switch-error count, adjacent switch pairs are counted as flips, and the Hamming distance is the minimum over the two orientations of misassigned variants. **b**, Block intersection and the block-level statistics. Only variants that are heterozygous in both files and phased in both are compared (dark bars); blocks left with a single variant are ignored (outline). Switch error rate and Hamming distance are computed over the intersected blocks; the phased fraction counts truth heterozygous variants matched and phased in the query before the partition, and the block N50 is computed on the query's own blocks of at least two variants, as implemented in `longphase compare`.

*Placeholder: panels to be regenerated with the submitted version.*

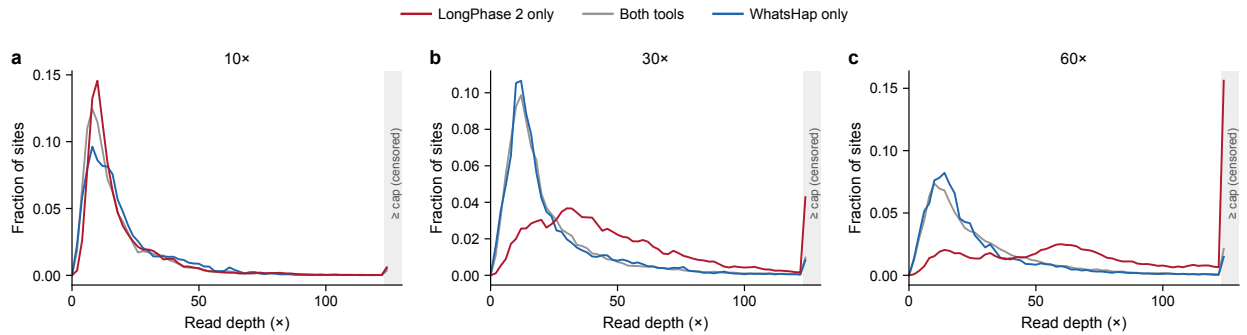
**Supplementary Fig. 9: Tuning of the voting and read-filtering parameters.** Sweeps on HG002 at 10–60 $\times$ ; the value adopted in LongPhase 2 is marked. **a**, Distribution of the pair ambiguity  $s$  over all assessed pairs and over pairs in a switch error. **b**, Switch errors and block N50 as a function of the no-vote threshold  $\beta$ . **c**, Switch errors as a function of the up-weight  $w$  applied when  $s \leq \alpha$ . **d**, Base-quality distribution over all bases and over erroneous bases, and switch errors as a function of the base-quality threshold  $\theta$ . **e**, Switch errors over the single-read-guard parameters  $N$  and  $\mu$ , and with and without the guard. [Placeholder: produce or remove. The five panels exist in the group's thesis material (Po-Hui Tzeng, 2026, Figs. 3.7, 3.10–3.12, 3.14, 3.17, 3.18, 3.21) and would be regenerated with the submitted version on the PEPPER-Margin-DeepVariant calls and coverages used here, in one consistent style; state two caveats: in **b** the switch-error curve has no interior optimum and  $\beta$  rests on the N50 trade-off, and in **e** the guard slightly *increased* switch errors at 10 $\times$ . If the figure is removed, delete the sentences citing it in Methods and Supplementary Method 3.]



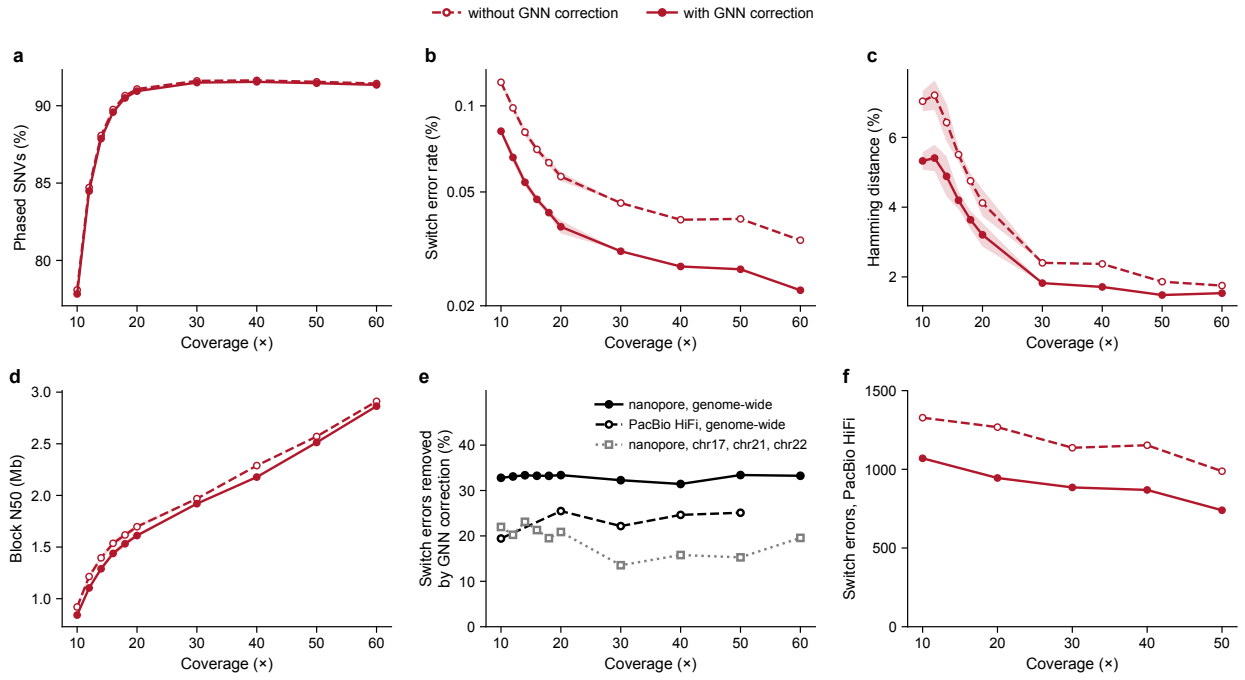
**Supplementary Fig. 10: SNV-only phasing compared with Margin and Ralphi.** HG002 nanopore R10.4.1 data at 10–20 $\times$ , means of ten replicates with s.d., scored against v5.0q; LongPhase 2 with GNN correction. **a**, Switch error rate. **b**, Hamming distance. **c**, Block N50; Ralphi is omitted because its blocks sum to more than the autosome length. **d**, Number of phase blocks. Values in Supplementary Table 8.



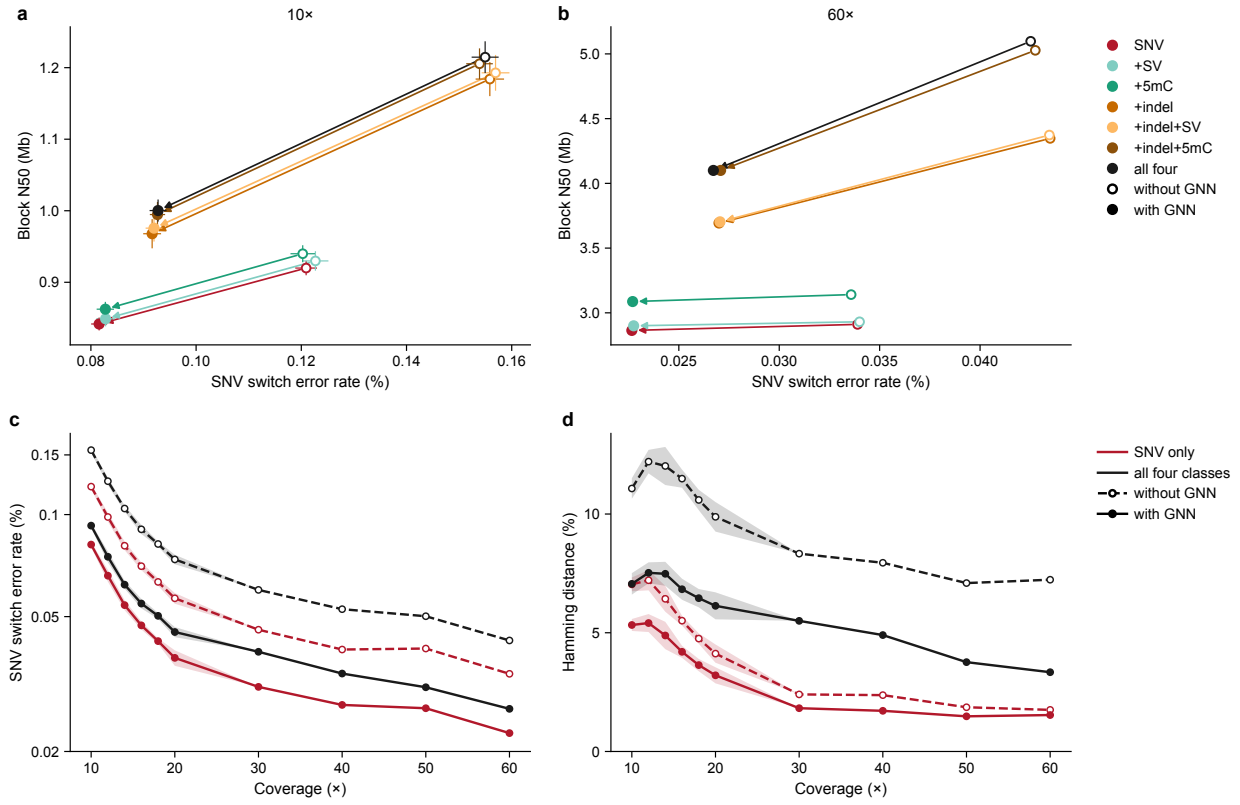
**Supplementary Fig. 11: Joint methylation co-phasing compared with MethPhaser.** HG002 nanopore R10.4.1 data, replicate 1, scored on SNVs against v5.0q. MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction. **a**, Switch errors. **b**, Fraction of benchmark heterozygous SNVs phased. **c**, Hamming distance. **d**, Block N50.



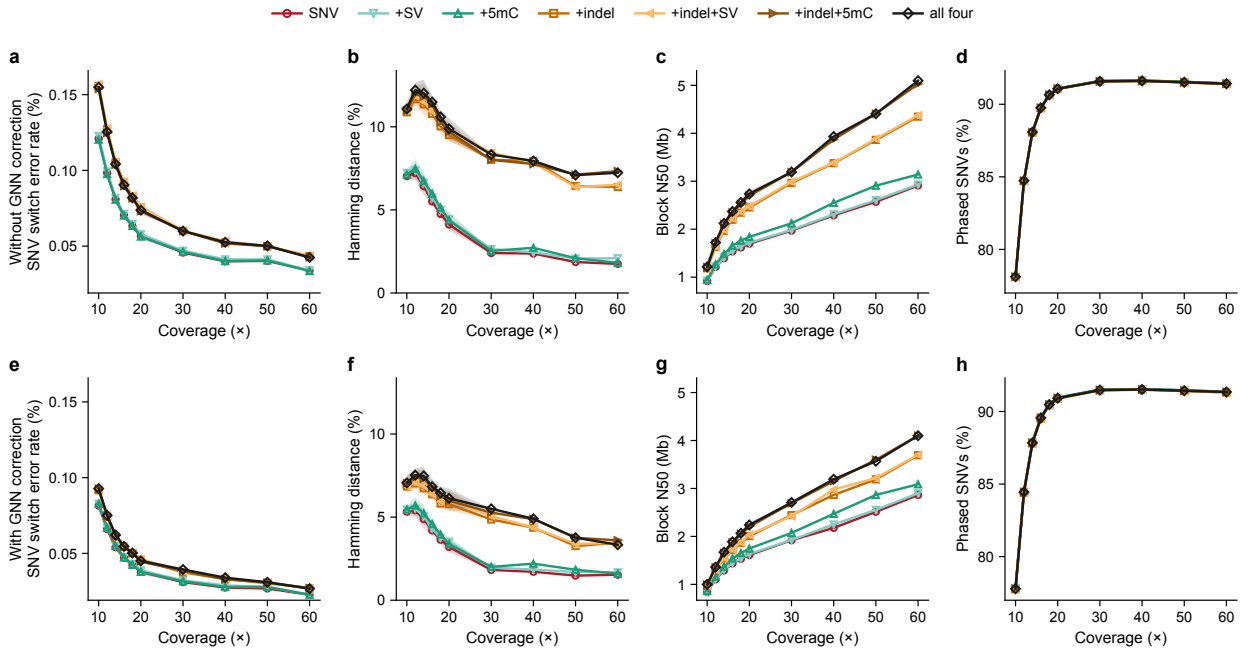
**Supplementary Fig. 12: Read depth at heterozygous SNV calls left unphased by LongPhase 2, by WhatsHap or by both.** HG002 nanopore R10.4.1 data, replicate 1, at 10 $\times$  (**a**), 30 $\times$  (**b**) and 60 $\times$  (**c**); 2 $\times$  bins. The grey bin collects every site at or above the depth cap of the variant caller (about 125 $\times$ ), whose true depth is censored. Sets as in Fig. 6a.



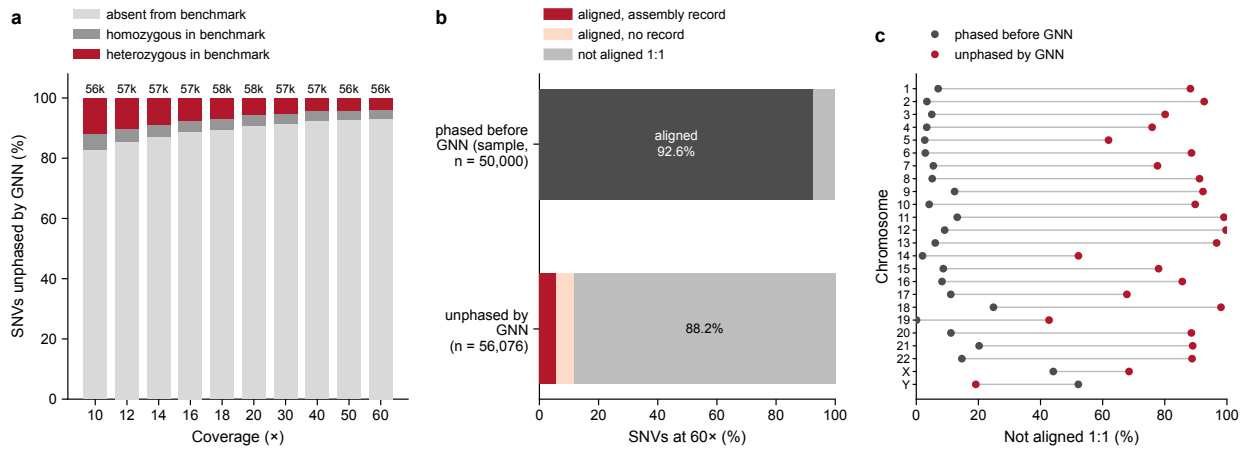
**Supplementary Fig. 13: Effect of GNN correction on SNV-only phasing.** LongPhase 2 with and without GNN correction; HG002 nanopore R10.4.1 data unless stated; means of ten replicates with s.d. at 10–20×; one down-sampling at 30–60×. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate. **c**, Hamming distance. **d**, Block N50. **e**, Fraction of switch errors removed by GNN correction on nanopore and PacBio HiFi data, genome-wide, and on the nanopore chromosomes held out from GNN training (chr17, chr21 and chr22; Supplementary Table 15). **f**, Switch errors on PacBio HiFi data, with the network trained on nanopore data.



**Supplementary Fig. 14: Co-phasing configurations with and without GNN correction.** HG002 nanopore R10.4.1 data; seven combinations of variant classes, each phased separately; SNV metrics against v5.0q. **a,b**, SNV switch error rate against block N50 at 10× (**a**; means of ten replicates with s.d.) and 60× (**b**); arrows run from the result without (open) to the result with (filled) GNN correction. **c,d**, SNV switch error rate (**c**) and Hamming distance (**d**; computed over SNVs and indels for four-class phasing) for SNV-only and four-class phasing.



**Supplementary Fig. 15: All co-phasing configurations across coverage.** HG002 nanopore R10.4.1 data; means of ten replicates with s.d. at 10–20×; one down-sampling at 30–60×. Top row without, bottom row with GNN correction. **a,e**, SNV switch error rate. **b,f**, Hamming distance (over SNVs and indels where indels are phased). **c,g**, Block N50. **d,h**, Fraction of benchmark heterozygous SNVs phased. Values in Supplementary Table 10.



**Supplementary Fig. 16: SNVs unphased by GNN correction.** HG002 nanopore R10.4.1 data, SNV-only phasing, replicate 1, final GNN model. **a**, Classification against the v5.0q benchmark VCF, matched by position without a benchmark BED; numbers above bars, total unphased. **b**, Status in the T2T-HG002 v1.1 assembly of the SNVs unphased at 60 $\times$  and of a random sample of SNVs phased before GNN correction: inside or outside the regions where both assembled haplotypes align 1:1 to GRCh38, and, inside, whether an assembly variant record starts at the same coordinate. **c**, Per chromosome, fraction of each set outside the 1:1 alignment. All chromosomes, including chrX and chrY; values and autosome-only shares in Supplementary Table 12.

## References

- [1] Rampášek, L. *et al.* Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*, vol. 35 (2022).
- [2] Veličković, P. *et al.* Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).
- [3] Brody, S., Alon, U. & Yahav, E. How attentive are graph attention networks? In *International Conference on Learning Representations (ICLR)* (2022).
- [4] Vaswani, A. *et al.* Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).
- [5] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)* (2017).
- [6] Patterson, M. *et al.* WhatsHap: weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology* **22**, 498–509 (2015).
- [7] Shafin, K. *et al.* Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nature Methods* **18**, 1322–1332 (2021).
- [8] Hansen, N. F. *et al.* A complete diploid human genome benchmark for personalized genomics. *Cell* **189**, 4857–4875.e31 (2026).
- [9] Olson, N. D. *et al.* The Genome in a Bottle HG002 assembly-based variant benchmark set enables comprehensive benchmarking of small and structural variants. *bioRxiv* (2026).
- [10] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).
- [11] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).
- [12] Zheng, Z. *et al.* Clair3-RNA: a deep learning-based small variant caller for long-read RNA sequencing data. *Nature Communications* **16** (2025).
- [13] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).
- [14] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).
- [15] Oxford Nanopore Technologies. epi2me-labs/wf-human-variation: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. <https://github.com/epi2me-labs/wf-human-variation> (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).
- [16] Pacific Biosciences. HiFi-somatic-WDL: tumor-normal variant calling workflow using HiFi reads. <https://github.com/PacificBiosciences/HiFi-Somatic-WDL> (2025).
- [17] Cao, S. *et al.* cuteHap: haplotype-aware structural variant detection in phased long-read sequencing data. *Advanced Science* **13** (2026).
- [18] Wang, S. *et al.* A scalable framework for comprehensive typing of polymorphic immune genes from long-read data. *Advanced Science* **13** (2026).
- [19] Gabbutt, C. *et al.* Fluctuating DNA methylation tracks cancer evolution at clinical scale. *Nature* **645**, 764–773 (2025).

- [20] O'Neill, K. *et al.* Long-read sequencing of an advanced cancer cohort resolves rearrangements, unravels haplotypes, and reveals methylation landscapes. *Cell Genomics* **4**, 100674 (2024).
- [21] LaFlamme, C. W. *et al.* Diagnostic utility of DNA methylation analysis in genetically unsolved pediatric epilepsies and CHD2 epismutation refinement. *Nature Communications* **15** (2024).
- [22] Iwasaki, Y. *et al.* Bidirectional disruption of *GNAS* transcripts causes broad methylation defects in pseudohypoparathyroidism type 1B. *Proceedings of the National Academy of Sciences* **122** (2025).
- [23] Zalusky, M. P. *et al.* 3-hour genome sequencing and targeted analysis to rapidly assess genetic risk. *Genetics in Medicine Open* **2**, 101833 (2024).
- [24] Smits, D. J. *et al.* Nanopore long-read sequencing for the critically ill facilitates ultrarapid diagnostics and urgent clinical decision making. *European Journal of Human Genetics* **34**, 108–118 (2025).
- [25] Fleming, A. *et al.* Combined approaches, including long-read sequencing, address the diagnostic challenge of HYDIN in primary ciliary dyskinesia. *European Journal of Human Genetics* **32**, 1074–1085 (2024).
- [26] Thomas, R. *et al.* Splice-modulating antisense oligonucleotides targeting a pathogenic intronic variant in adult polyglucosan body disease correct mis-splicing and restore enzyme activity in patient cells. *Nucleic Acids Research* **53** (2025).
- [27] Tian, Y., McDonnell, S. K., Wu, L., Larson, N. B. & Wang, L. Fine mapping regulatory variants by characterizing native CpG methylation with nanopore long-read sequencing. *Human Genetics and Genomics Advances* **7**, 100532 (2026).
- [28] Galvez, J. H. *et al.* Benchmarking of Oxford Nanopore whole genome sequencing for germline variant and CpG methylation detection across Canada's national platform for genome sequencing and analysis. *Frontiers in Genetics* **17** (2026).
- [29] Flack, N. *et al.* Chromosome-level, nanopore-only genome and allele-specific DNA methylation of Pallas's cat, *Otocolobus manul*. *NAR Genomics and Bioinformatics* **5** (2023).
- [30] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).
- [31] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).
- [32] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).
- [33] Zhou, Y., Leung, A. W.-S., Ahmed, S. S., Lam, T.-W. & Luo, R. Duet: SNP-assisted structural variant calling and phasing using Oxford nanopore sequencing. *BMC Bioinformatics* **23** (2022).